Are Frontier VLM Agents Ready to Be Robot Generalists?

An Empirical Study with the Embodied Agent Arena

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Abstract. Frontier vision-language models (VLMs) increasingly estimate scenes, ground interactions, and generate executable actions. How far these native capabilities support embodied generalism across diverse tasks remains unclear. We introduce Embodied Agent Arena to assess nine VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation. The arena contains 1,000 cases drawn from 32 established sources and GeoProbe, our new benchmark for geometric estimation on Blender renders and real-scene images. Models make task decisions through a minimal harness that preserves source observations and operations. Separate measures of metric precision, functional grounding, and native goal completion connect local competence to complete task outcomes. Astra has the lowest pose errors and highest full-set planning success, while Opus 5.5 achieves the highest affordance overlap, contact validity, and manipulation success. DeepSeek-1017 completes 20.9% of Manipulation tasks, with spatial and planning performance varying sharply across sources. Supplementary experiments show that per-round visual feedback and multi-round review can change model rankings. Current VLM agents still fall short of embodied generalism, leaving task goals unsatisfied within finite time and interaction budgets.

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1 Introduction

A robot generalist must turn scene understanding into actions that satisfy a complete goal. Locating an object, selecting a usable contact, and executing a grasp do not establish that the object reaches its destination. Vision-language models (VLMs) increasingly address spatial relations (Ma et al., 2024, Yang et al., 2024) and action-relevant grounding (Wang et al., 2025a, Song et al., 2024). Learned visuomotor policies connect observations to robot actions (Kim et al., 2024, Black et al., 2024), while language-model controllers select skills or generate executable programs (Ahn et al., 2022, Liang et al., 2022). Frontier agents increasingly combine these mechanisms with perception support and visual action rehearsal (Fu et al., 2026, Guo et al., 2026, Li et al., 2026b, Zhang et al., 2026a). This progress raises a central issue: which robotic requirements can the VLM's own perception, reasoning, and action selection satisfy?

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Figure 1. Overview of Embodied Agent Arena. Each row illustrates six tasks in one capability domain, spanning camera and scene estimation, spatial relations, functional grounding, goal-directed planning, and robot manipulation. Fixed-input probes test estimates and judgments; interactive episodes test goal completion. Spatial-tracing overlays show reference paths and a 3D target region.

Assessing generalism requires connecting individual capabilities to complete task performance. Aggregate scores can hide errors in metric scale, reference frames, or placement conditions; terminal failure can conceal useful progress. This distinction motivates our central question: *Are frontier VLM agents ready to be robot generalists?*

We introduce Embodied Agent Arena to evaluate nine VLM agents across Geometry, Spatial Reasoning, Affordance, Task Planning, and Manipulation (Figure 1). Its 1,000 cases combine 32 established sources with GeoProbe, which varies camera motion, object motion, and scale references to diagnose geometric estimation. A minimal harness retains each source's observations and operations while leaving task decisions to the model. Separate measures of geometric precision, functional grounding, and native task completion reveal both cross-capability strengths and the requirements left unsatisfied within a task.

The results expose a gap between local precision and complete execution: Astra produces the closest paths yet often misses their endpoints, and its local control strengths coexist with low household-goal completion. Supplementary experiments examine richer observations and multi-round review, showing that additional evidence has model- and task-dependent effects. Our contributions are:

- **An evaluation perspective centered on task requirements.** We distinguish local estimates and interaction evidence from complete goal satisfaction across five domains.

- **A unified arena with a new geometric benchmark, GeoProbe.** We combine 32 established sources with 168 GeoProbe cases, using controlled Blender scenes and real images to diagnose camera, motion, depth, and scale estimation.
- **A diagnosis of the gap from local competence to complete tasks.** We compare nine agents and identify the task requirements that remain unsatisfied despite strong geometric precision and functional-contact localization.

2 Related Work

Embodied evaluation has developed along two complementary directions. Control suites test execution and task generalization (James et al., 2019, Mees et al., 2021), whereas visual diagnostics isolate spatial inference (Ma et al., 2024, Yang et al., 2024) and interaction grounding (Song et al., 2024, Qian & Fouhey, 2023). Scene and task question answering (Majumdar et al., 2024, Sermanet et al., 2023) extends these diagnostics to embodied decisions (Li et al., 2024b, Qi et al., 2025). The former establishes whether a goal is achieved; the latter resolves particular perceptual or reasoning requirements. Evaluating both in the same agents connects local competence to complete task outcomes.

The interface determines how that competence reaches execution. Learned policies map observations and instructions to actions (Brohan et al., 2023, Zitkovich et al., 2023), increasingly across tasks (Ghosh et al., 2024, Kim et al., 2024, Black et al., 2024); language-mediated control instead selects supplied skills (Ahn et al., 2022, Huang et al., 2023b) or composes executable programs (Liang et al., 2022, Singh et al., 2023). Programs organize control primitives and dependencies (Fu et al., 2026, Du et al., 2025), geometric feedback (Huang et al., 2023a), and reusable policy interfaces (Li et al., 2026a). Their decisions still depend on visual evidence quality (Li et al., 2025, Wu et al., 2026) and temporal consistency (Huang et al., 2026b, Zhao et al., 2026b,a). Demonstrations (Jiang et al., 2023, Huang et al., 2026a), retained histories (Zhang et al., 2026b), and execution verification (Guo et al., 2024, Huang et al., 2026c) address these dependencies. Task success therefore depends on model decisions and interface support.

Broader evaluations connect perception with goal-directed interaction (Yang et al., 2025a, Cheng et al., 2025). Table 1 compares their coverage, diagnostics, and scale: OpenEQA (Majumdar et al., 2024), Embodied Arena, and Hard Vision, Easy Vision combine visual probes with navigation (Ni et al., 2025, Rasheed et al., 2026); VLABench also evaluates manipulation (Zhang et al., 2024). Embodied Agent Arena connects geometry, functional grounding, and native task completion across 1,000 cases, with GeoProbe isolating camera, motion, and scale requirements. This unified evaluation measures the current level of VLM agents' native embodied generalist capabilities across tasks, from local judgments to task completion.

3 The Embodied Agent Arena

3.1 Benchmark Construction and Capability Domains

Data construction. Embodied Agent Arena contains 1,000 cases from 32 established sources and one new benchmark, GeoProbe, organized into five capability domains (Figures 1–2). Each case specifies observations, a query or goal, operations, and scoring. We introduce GeoProbe to diagnose the metric ambiguities behind geometric estimates. Its 168 cases vary camera motion, object motion, and scale references in controlled Blender scenes, complemented by

Table 1. Evaluation coverage, diagnostic analysis, and sample scale. Fixed probes use supplied observations; region probes score image grounding; manipulation requires arm/gripper control; memory spans sessions. Calib./pose tests camera calibration or relative pose; Forms denotes task-type/condition breakdowns. Scale counts QA items or task episodes; NR: aggregate count not reported (Appendix G).

	Sample scale	Evaluation coverage					Diagnostic analysis	
Benchmark	Instances	Fixed	Region	Nav.	Manip.	Memory	Calib./pose	Forms
OpenEQA (2024)	1,636	✓	✗	✓	✗	✗	✗	✓
VLABench (2024)	NR	✓	✗	✗	✓	✗	✗	✓
BEAR (2025)	4,469	✓	✓	✗	✓*	✗	✗	✓
EmbodiedBench (2025a)	1,128	✗	✗	✓	✓	✗	✗	✓
EmbodiedEval (2025)	328	✗	✗	✓	✗	✗	✗	✓
Embodied Arena (2025)	NR	✓	✓	✓	✗	✗	✗	✓
CaP-X (2026)	700/tier	✗	✗	✓	✓	✗	✗	✓
WorldLines (2026b)	331	✓	✗	✗	✗	✓	✗	✓
Hard/Easy Vision (2026)	NR	✓	✓	✓	✗	✗	✗	✓
Embodied Agent Arena	**1,000**	✓	✓	✓	✓	✗	✓	✓

✓ included; ✗ not evaluated; *BEAR-Agent online validation. Scale details: Appendix G.

real images. For interactive cases, the recorded terminal state connects the agent's action sequence to the task's goal conditions. Appendix A gives source allocation and adaptations.

Geometry. Given images, the agent estimates intrinsics, pose, depth, dimensions, or displacement. InFlux and Map-free test calibration and relative pose (Liang et al., 2025, Arnold et al., 2022); GeoProbe varies camera motion and scale references. TraceSpatial-Bench adds paths on 56 scenes (Zhou et al., 2026), testing reference-trajectory error alongside start, arrival, and collision constraints. These targets separate numerical precision from geometric feasibility.

Spatial Reasoning. The agent infers relations, viewpoints, configurations, and temporal order from images or video (Ma et al., 2024, Wang et al., 2025b, Yang et al., 2024). Examples include comparing object distances, imagining another observer's view, and inferring camera motion. Each question specifies a reference frame and expects a category, quantity, or point set.

Affordance. Given a scene and intended action, the agent predicts a contact point or bounding box. UMD tests tool functions such as cutting and scooping (Myers et al., 2015); instruction- and action-conditioned 3DOI queries test interaction contacts (Qian & Fouhey, 2023). Point validity tests whether the contact lies in the functional region; overlap measures region coverage.

Task Planning. The agent organizes source-provided operations into a sequence that achieves a household, navigation, or scientific goal (Shridhar et al., 2019, Wang et al., 2022), using visual, textual, or symbolic feedback. Compound goals require several terminal conditions, such as heating an object and placing it at the destination. For terminal state s_T and required predicates $g_1, \ldots, g_K$, complete success is

$$\mathrm{Success}(s_T, g) = \mathbf{1}\left[\bigwedge_{k=1}^{K} g_k(s_T)\right]. \tag{1}$$

Manipulation. The agent executes robot motions or skills for placement, insertion, articulation, and coordinated handling (James et al., 2019, Chen et al., 2025a). Language-conditioned variants change the target relation or object (Zhang et al., 2024); sequential tasks require preserving

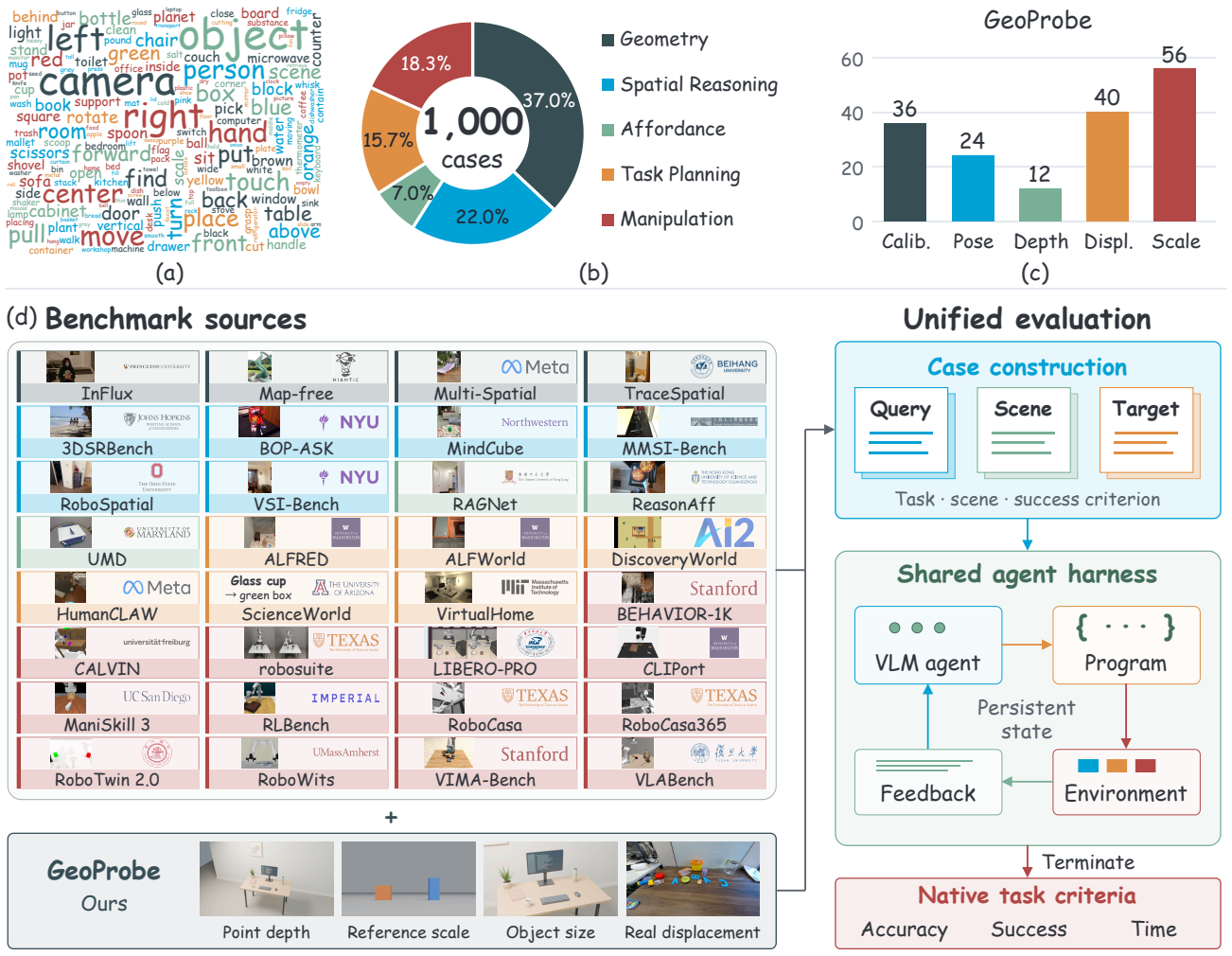

Figure 2. Construction of the 1,000-case arena. (a) Representative task vocabulary across sources. (b) Five-domain composition, including 112 spatial-tracing cases in Geometry. (c) GeoProbe estimation targets. (d) Source queries, scenes, and targets connect model interaction to termination and scoring. Colors identify domains; logos denote source institutions.

progress across ordered goals. Native success checks task completion, while condition-level scores record achieved relations and control events.

3.2 Unified Agent Harness

Model-directed interaction. The harness supports persistent model-directed interaction (Figure 2d). Geometry, Spatial Reasoning, Affordance, and Planning use a Python runtime; Manipulation follows its native agent loop. Models interpret observations, select operations, and construct action sequences using the resulting feedback. Here, native capability means model-directed task decisions through the supplied source interfaces, including their helpers. The baseline adds no perception or control modules beyond those interfaces; the supplementary review study adds specialist measurements. Variables persist across Python rounds, and images produced during a round enter the next model request. Appendix B.1 details the interfaces; Section 4.4 studies tool-assisted review (Zhang et al., 2025, Huang et al., 2026c).

Submission and termination. Fixed-input submissions commit predictions; interactive runs end on agent termination, an environment terminal condition, or budget exhaustion. Scoring independently checks outputs or states. Appendix B.2 details budgets.

3.3 Evaluation Protocol and Metrics

① *Task success.* Spatial Reasoning and Planning report Pass@1. Manipulation reports native success, consecutive goals completed, and task-specific condition or control-evidence scores. Categorical spatial accuracy is reported separately. ② *Geometric precision.* Geometry and numerical spatial queries use MAE, RMSE, AbsRel/MAPE, displacement error, and pose errors according to the target. Tracing pairs trajectory RMSE with offline 3D constraint satisfaction. ③ *Native affordance grounding.* The main table reports box–mask IoU for native rectangular outputs and polygon–mask IoU for a separate direct-polygon experiment on the same cases. Both compare rasterized regions with the reference mask. Contact validity, box–box IoU, and strict box success remain complementary diagnostics in the appendix. ④ *Variation and resource use.* Repeated binary rates report their mean and standard deviation (SD) across runs. Continuous metrics pool valid case–run observations rather than averaging run means. Tokens measure inference use; action calls measure executed control. Section 4.1 specifies repetitions; Appendix B.3 defines aggregation.

4 Experiments

4.1 Experimental Setup

① *Models and configuration.* We evaluate nine VLM agents: Astra, Sol, Fable, Opus 5.5, Qwen-Max, Gemini, Qwen-397B, Qwen-27B, and DeepSeek-V4.1-Flash-1017 (DeepSeek-1017). Appendix B.2 records identifiers and source-specific limits. The DeepSeek-1017 endpoint is `deepseek-flash`, evaluated after its 17 October 2026 update. ② *Evaluation and aggregation.* In the original cohort, non-tracing Geometry, Spatial Reasoning, and Affordance use five runs per case, or three for Fable. DeepSeek-1017 follows the five-run schedule. Opus 5.5 uses five runs on non-tracing Geometry and one elsewhere. Tracing and execution tasks use one episode per assignment. Continuous metrics pool valid observations; rates retain assigned cases. Planning covers all 157 cases per model. Manipulation retains all 182 assignments, counting two Opus 5.5 RLBench errors and one RoboTwin error as unsuccessful. All agents use native RGB observations and tools. The original eight agents follow the Baseline observation schedule; DeepSeek-1017 receives current RGB at each decision. Their per-source execution ceilings are listed in Appendices E.1 and C.4. The direct-polygon experiment uses one run on the same 70 Affordance cases for all nine models (Appendix C.5); Appendix C.3 details Opus 5.5 repetitions.

4.2 Main Results

Local strengths differ across models and metrics. Astra has the lowest pooled pose errors (17.1°, 106.6 cm) and highest full-allocation Planning success (80.3%). Its Spatial Pass@1 narrowly exceeds Qwen-Max's (69.4% versus 68.9%), whereas Gemini has lower numerical AbsRel (0.161 versus 0.213). Opus 5.5 leads both box–mask IoU (0.329) and direct-polygon IoU (0.397), versus Astra's 0.286 and 0.379. Figure 3 summarizes the resulting capability profiles.

Local accuracy does not establish complete execution. Opus 5.5 reaches the highest Manipulation success at 47.8%, versus Astra's 41.8%, yet even the leading model leaves more than

Table 2. Performance across five domains. Box–mask and Polygon IoU use separate output protocols on the same 70 cases. Planning covers 157 cases and Manipulation 182 assignments. ± denotes run SD; best means are bold.

	Geometry		Spatial Reasoning		Affordance		Task Planning	Manipulation
Agent	Rot.↓ (°)	Trans.↓ (cm)	Pass@1↑ (%)	AbsRel↓	Box–mask↑ IoU	Polygon↑ IoU	Pass@1↑ (%)	Task SR↑ (%)
Astra	**17.1**	**106.6**	**69**.4±1.0	0.213	0.286	0.379	**80.3**	41.8
Sol	27.0	151.9	60.7±2.5	0.273	0.264	0.325	44.6	19.8
Fable	25.2	121.7	62.0±4.6	0.186	0.284	0.328	63.7	24.7
Qwen-Max	20.5	138.1	68.9±1.1	0.205	0.272	0.365	35.7	9.3
Gemini	24.2	140.9	65.4±1.3	**0.161**	0.250	0.291	48.4	18.7
Qwen-397B	38.4	249.8	50.6±1.8	0.251	0.116	0.092	17.8	9.9
Qwen-27B	38.4	213.7	48.0±0.9	0.288	0.013	0.051	19.1	11.0
Opus 5.5	23.3	135.3	58.6	0.222	**0.329**	**0.397**	48.4	**47.8**
DeepSeek-1017	30.3	182.7	51.6±1.5	0.310	0.217	0.249	31.2	20.9

half of the assigned tasks incomplete. DeepSeek-1017 solves 49/157 Planning tasks (31.2%) and 38/182 Manipulation tasks (20.9%). All nine models fail to complete the five BEHAVIOR-1K tasks: their recorded final states leave required task conditions unsatisfied within the execution limits. The following diagnostics locate where local competence stops short of complete execution; Section 4.4 examines the effect of per-round RGB under matched budgets.

4.3 A Deep Dive of Astra

Figures 4–8 trace Astra's local strengths through to task outcomes. Captions identify each diagnostic's models; Appendices D, C.3, and C.4 give the task-level results.

Geometry. ① *Astra leads controlled metric estimation across targets.* Among the original seven agents, Astra has the lowest errors on all five Blender estimation targets (Figure 4a), including reference height at 0.31% versus Sol's 0.90%. Depth ordering reaches 96.3%, while oracle scale alignment lowers depth AbsRel from 0.104 to 0.038: much of its remaining depth error concerns scale rather than order. InFlux focal error reverses the ranking (53.0% versus Sol's 47.8%), locating the strongest advantage in controlled geometric cues. ② *Astra separates object displacement from moderate viewpoint change.* Across fixed, small-shift, and large-shift Blender conditions, Astra's MAE is 0.24, 0.57, and 1.13 cm; Sol is the closest original peer throughout (Figure 4b). The fixed-camera lead survives excluding stationary objects. On four real tabletop photo pairs, Astra achieves 0.05 versus Sol's 0.42 cm MAE. The lower real-scene error occurs on different scenes and object motions. ③ *Coupled motion challenges Astra's otherwise leading pose estimates.* Its aggregate pose errors are lowest (Table 2), yet none of the original seven agents meets both rotation and translation tolerances under mixed motion. ④ *Astra traces the closest paths, but endpoints decide success.* Astra has the lowest 3D trajectory RMSE at 0.382 m versus Gemini's 0.426, whereas joint success is 33.9% versus 37.5% (Figure 4c). Opus 5.5 records 0.683 m and 8.9% joint success; DeepSeek-1017 records 0.470 m and 14.3%. Initiation or arrival fails in 36 of Astra's 37 unsuccessful traces, identifying endpoint placement as the main obstacle to successful tracing.

Spatial Reasoning. ① *Astra's clearest advantage is relating multiple views.* Across all MMSI tasks it reaches 77.0% versus Fable's 66.7%; across MindCube it ties Qwen-Max at 90.8%. Within these sources, object relations reach 96.7% and camera–region relations 100% (Figure 5a; Appendix D.2). Opus 5.5 reaches 74% on MindCube and 45% on MMSI, while its 95% BOP-ASK accuracy approaches Astra's 96% (Figure 5c). DeepSeek-1017 reaches 88.0% on BOP-ASK but 20.0% on VSI-Bench, reinforcing the source-dependent pattern. In the original cohort, equal weighting of all six sources also ranks Astra first, at 71.9% versus Qwen-Max's 69.1%

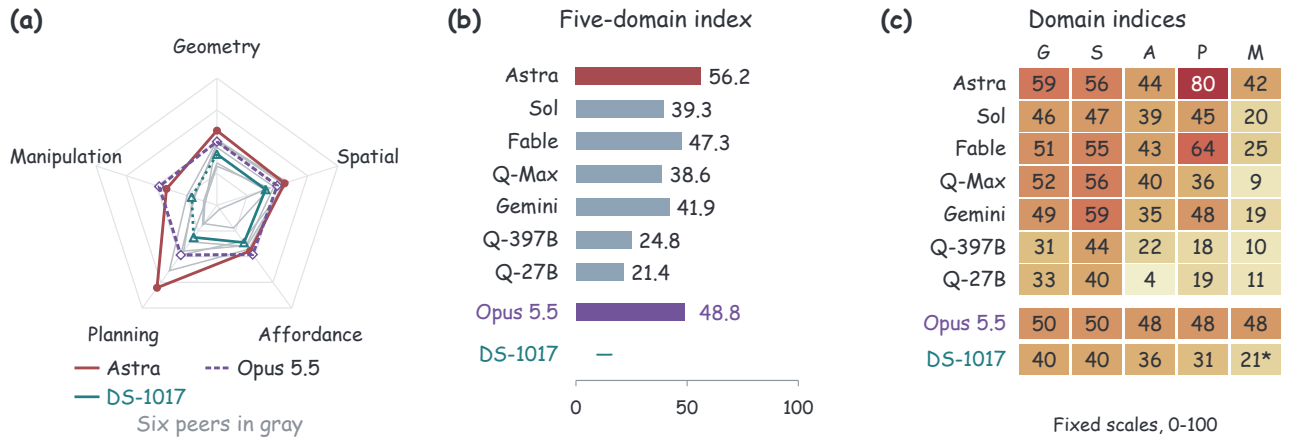

Figure 3. Capability breadth across nine agents. (a) Astra (red), Opus 5.5 (purple), DeepSeek-1017 (DS-1017, teal), and six peers (gray). (b) Mean of five domain indices for the eight Baseline agents. (c) Indices by domain; *denotes per-round RGB in Manipulation. Appendix C.6 defines the scales. Planning includes all 157 cases. Affordance combines box–mask IoU and contact validity; Table 2 also reports polygon IoU.

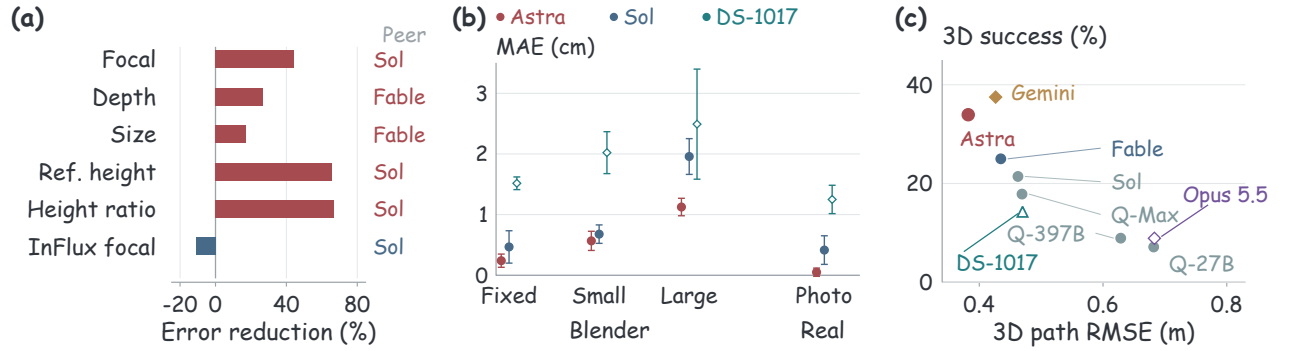

Figure 4. Geometric precision and task completion. (a) Astra's error reduction relative to the lowest-error original peer; the first five targets are GeoProbe. (b) Displacement under fixed, small-shift, and large-shift Blender cameras, plus separate real photos; mean ± run SD. Panel a retains the original seven-agent diagnostic; b adds DeepSeek-1017 under the same camera conditions. (c) Single-run 3D tracing error versus joint success for all nine agents. Appendix D.1 gives task-level comparisons; Appendices C.3 and C.4 detail the added models.

(Appendix C.2). ② *Astra's strength in relations outpaces its inference of new frames.* MindCube camera-shift accuracy falls to 78.5% versus Qwen-Max's 89.2%; 3DSRBench object-facing judgments reach 56.7% versus Fable's 72.2% (Figure 5a). The shared-run comparisons in Appendix D.2 preserve both deficits. Relating visible landmarks and recovering motion or an object's own axes make different demands. ③ *Depth cues are more reliable than choosing among objects.* Astra achieves 96–100% on BOP-ASK camera-depth ordering, but 50% on VSI nearest-object selection versus Qwen-Max's 76.7% (Figure 5b). RoboSpatial configuration is also 22 points behind its best original peer (Appendix D.2). Thus its narrow overall Spatial lead combines strong multi-view relations with unresolved object-centered selection and configuration.

Affordance. ① *Astra leads contact localization in the original cohort.* It leads UMD and ReasonAff at 96.7% and 31.0%, versus Fable's 94.4% and 23.3%, and ties Fable at 35.0% on RAGNet. DeepSeek-1017 reaches 82.7% UMD contact validity (Figure 6a). ② *Native boxes locate usable parts without reproducing their shape.* Cutting reaches 100% contact validity and 96% strict box

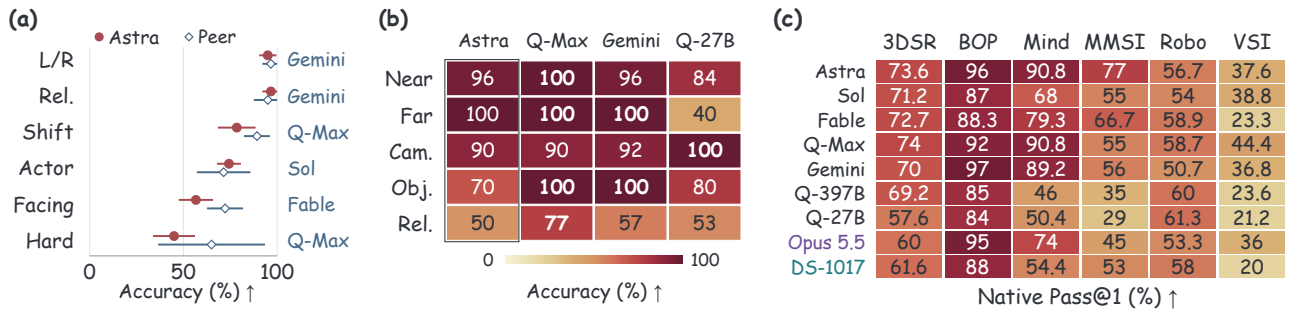

Figure 5. Spatial strengths depend on task and source. (a) Astra's relation and viewpoint accuracy versus the best original peer. (b) Camera-relative depth (Near/Far/Cam.) and inter-object distance (Obj./Rel.). Panels a–b retain the original seven-agent diagnostics and run SDs. (c) Native Pass@1 for nine agents across six sources, including numerical questions; 3DSR, Mind, Robo, and VSI abbreviate 3DSRBench, MindCube, RoboSpatial-Home, and VSI-Bench. Opus 5.5 uses one run and DeepSeek-1017 five.

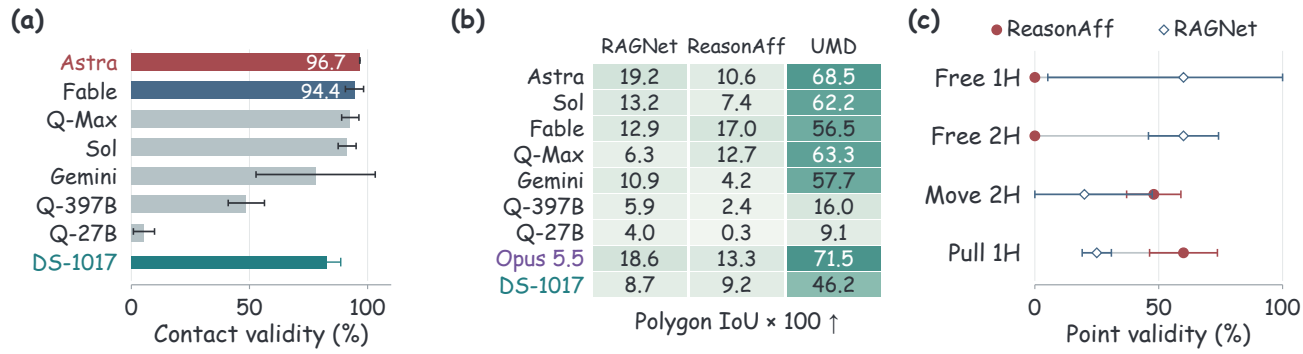

Figure 6. Contact localization and region shape reveal different demands. (a) UMD contact validity, with DeepSeek-1017 added to the original seven-model diagnostic. (b) Direct-polygon IoU (%) for all nine models, one run per case; invalid or absent polygons score zero. (c) Astra's original contact validity for ReasonAff/RAGNet actions; 1H/2H denotes one/two hands. Whiskers in a and c show run SD. Panel b uses a separate polygon-output protocol (Appendix C.5).

success, despite box–mask IoU of 0.361; box–box IoU is 0.694 (Appendix D.3). The added direct-polygon experiment tests shape without an external segmenter (Figure 6b). Opus 5.5 reaches 71.5% polygon IoU on UMD but only 18.6% on RAGNet and 13.3% on ReasonAff-style images; DeepSeek-1017 shows the same source disparity at 46.2%, 8.7%, and 9.2%, respectively. Flexible output geometry alone does not resolve interaction-conditioned grounding. Containment still has only 44% strict box success versus Qwen-Max's 96%, showing that contact localization and region extent are distinct even under the same output format. ③ *Action-conditioned contacts expose where Astra's lead changes.* ReasonAff pulling and two-handed movement reach 60% and 48%, above the best peers' 50% and 26.7%; the RAGNet counterparts reach 25% and 20%, below 31.7% and 50% (Figure 6c). The same panel shows zero success on both ReasonAff free-movement categories despite nonzero RAGNet scores. Contact choice therefore depends on the action–scene pairing, beyond recognizing a tool's functional part.

Task Planning. ① *Astra's planning lead spans all six sources.* Within the original cohort, it leads four sources and ties the best model on two (Figure 7a; Appendix D.4). ALFRED and ALFWorld success is 75% and 70%, versus Fable's 35% and 30% and Opus 5.5's 15% and 45%; DeepSeek-1017 reaches 5% and 20%. Among the original agents, Astra alone solves ALFRED heating and dual placement (Figure 7b), extending its advantage to state changes and maintaining progress across multiple objects. ② *Astra gains most when scientific goals require linked operations.*

DiscoveryWorld reaches 100% versus Fable's 70%; ScienceWorld reaches 87.7% versus 80.8%. Figure 7c locates the largest gains in unknown-plant genetics and melting-point measurement. Thermometer use reverses this lead at 66.7% versus 83.3%, identifying instrument operation as a distinct bottleneck within scientific planning. ③ *Astra finds seating targets but fails to establish the sitting state.* It ties Gemini for the best HumanCLAW contact rate (60%), yet all nine agents fail all eight sitting tasks under their recorded budgets. Five Astra episodes reach the seat and issue the action; Opus 5.5 finds the target and approaches within 20 cm in four, without satisfying the sitting condition. Fable records no target-finding success in these episodes. This separates reaching a target from achieving the required interaction state.

Manipulation. ① *Astra's baseline lead spans several operation families.* Among the original seven agents, it leads eight of thirteen suites (Figure 8a). Examples include CLIPort at 61.1% versus Fable's 33.3%, RLBench at 60% versus Sol's 20%, and RoboTwin at 38% versus Fable's 14%, spanning placement, articulation, and coordinated handling. Opus 5.5 ties Astra on CLIPort and ManiSkill, exceeds it on RoboTwin (46% versus 38%), and trails on RLBench (30% versus 60%). Both agents complete five of nine ManiSkill tasks and attain a 60% condition score on the diagnostic subset (Figure 8c). ② *Astra converts baseline control into longer goal sequences.* Astra completes 3.10 consecutive CALVIN goals on average versus Gemini's 1.80, Sol's 1.40, and Fable's 0.90 (Figure 8b). Opus 5.5 averages 2.40 consecutive goals and completes one full sequence; Astra completes three of ten. None of the other original agents completes a full Baseline sequence. DeepSeek-1017 averages 1.40 goals and completes one full sequence. Panels b–c also show the paired observation experiment analyzed in Section 4.4. ③ *Household execution exceeds Astra's local control strengths.* Within the source-specific execution budgets, Astra completes one of ten RoboCasa tasks and none in RoboCasa365; all nine agents fail the five BEHAVIOR-1K tasks (Figure 8a). These tasks combine mobile-base positioning and articulated fixtures with object handling (Nasiriany et al., 2024). Household success requires coordinating navigation, object placement, and environment-state changes.

4.4 Further Investigations

Per-round RGB under matched budgets. We compare the original seven agents with their native observation schedule and with current RGB supplied at every decision. ① *Astra's control lead can reverse.* With inference budgets and execution-step limits fixed, per-round RGB changes CALVIN success from 30% to zero for Astra and zero to 40% for Fable (Table 3b); average consecutive-goal scores also reverse (Figure 8b). ② *Astra processes more input while executing fewer actions.* Astra's median input tokens rise from 0.32M to 1.21M, while action calls fall from 1,795 to 481. In sequence 0557, its reusable controller appears at round 12 rather than 5; expanded Fable builds one at round 2. This trajectory contrast links completion to when agents establish reusable feedback control (Appendix E.1). ③ *Stable totals hide opposing task effects.* VIMA improves for all three models, and RoboTwin improves for Astra. Across 153 common tasks, Astra stays at 63 successes while Fable rises from 34 to 65. ManiSkill condition scores likewise fall from 60% to zero for Astra, while expanded Fable reaches 60% (Figure 8c). Task-specific gains offset losses in Astra's aggregate.

Multi-round review on fixed inputs. Agents revisit initial answers on 46 shared cases. Self-review and Crop review allow two additional responses; Tool-assisted review combines six additional responses with specialist measurements (Appendix E.2). Tool-assisted review raises Astra's Affordance success from 22.2% to 33.3%, but improves five and worsens four Geometry entries (Table 3a). Astra's Spatial accuracy remains at 72.2%, while Qwen-27B rises from 55.6% to 61.1%. Qwen-27B's self-review yields higher box IoU than tool assistance (0.1524 versus

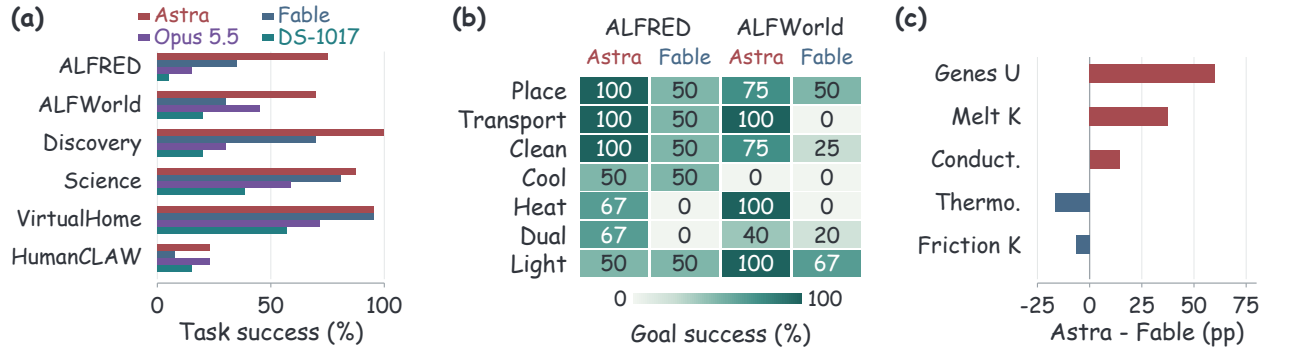

Figure 7. Planning success varies across sources and goals. (a) Source-level success for Astra, Fable, Opus 5.5, and DeepSeek-1017. DeepSeek-1017 completes 2/13 HumanCLAW tasks. Opus 5.5 completes 3/13 HumanCLAW tasks, all navigation; none of its eight sitting tasks succeeds. (b) Astra/Fable success on seven household goal families. (c) Their five nonzero ScienceWorld gaps; thirteen families tie, and K/U denotes known/unknown conditions. Each assignment uses one run. Appendix D.4 gives the task-family breakdown; Table A11 gives Opus 5.5 coverage.

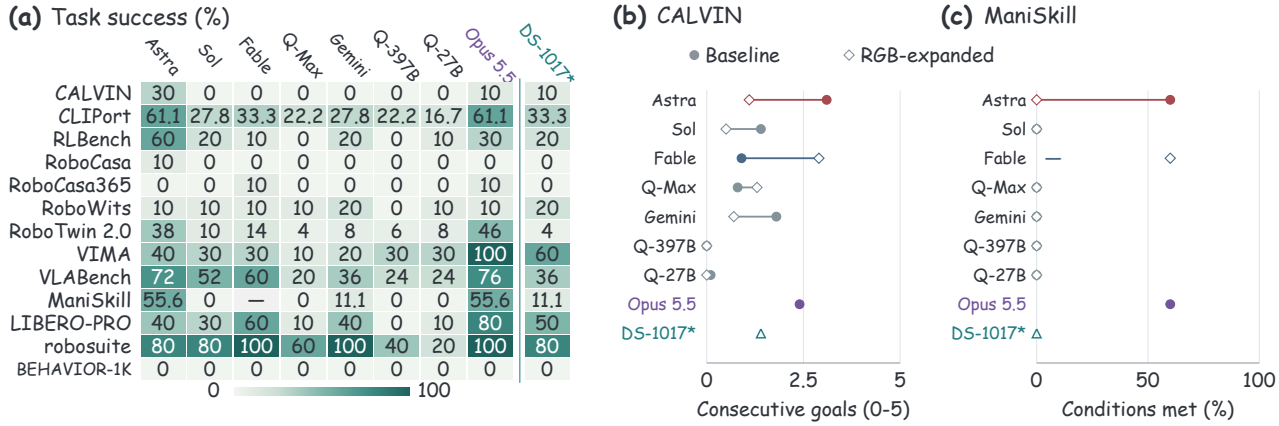

Figure 8. Manipulation across suites and execution settings. (a) Thirteen-suite success for nine agents. (b) Mean consecutive CALVIN goals, out of five. (c) Mean native condition satisfaction on five ManiSkill diagnostic tasks. Filled/open markers denote Baseline/RGB-expanded settings; teal triangles (*) show DeepSeek-1017 with per-round RGB. — marks an unavailable Fable result. Each task uses one run; execution errors count as failures. Appendices D.5, C.3, and C.4 give source results and settings.

0.1452), yet tool assistance alone produces a box passing the 0.5 threshold. Mean overlap and threshold success capture different effects of review. The tool-assisted condition measures the joint effect of added measurements and a larger inference budget.

Case studies. Appendix F connects spatial judgments and contact localization to controller construction and task completion in CALVIN and PickCube. Appendix H extends these examples across five capabilities and nine agents. Shared-task records (Table A39) separate completion, interaction use, and runtime: fewer rounds need not mean faster execution.

Together, these cases connect local model decisions to the conditions left unsatisfied at termination. Appendix E.3 relates these outcomes to task difficulty and external execution support.

Table 3. Multi-round review (a) and RGB-expanded Manipulation (b). Baseline and RGB-expanded share budgets and execution-step limits; RGB-expanded supplies per-round images. I/T/W counts improved/tied/worsened entries; SR_{50} requires box IoU≥ 0.5. Goals gives average consecutive completions (0–5); Common pools nine suites (153 tasks). Details: Appendix E.

(a)		Geometry	Spatial Reasoning	Affordance	
Model	Condition	I/T/W	Acc. (%)↑	Box IoU↑	SR_{50} (%)↑
Astra	Initial	–	**72.2**	0.2903	22.2
	Self-review	3/20/1	66.7	0.2914	22.2
	Crop review	4/19/1	66.7	0.2912	22.2
	Tool-assisted	5/15/4	**72.2**	**0.3174**	**33.3**
Qwen-27B	Initial	–	55.6	0.0336	0.0
	Self-review	6/15/3	50.0	**0.1524**	0.0
	Crop review	3/19/1	44.4	0.0937	0.0
	Tool-assisted	10/6/8	**61.1**	0.1452	**11.1**

(b)		CALVIN		ManiSkill	VIMA	Common
Model	Condition	SR (%)↑	Goals↑	SR (%)↑	SR (%)↑	SR (%)↑
Astra	Baseline	**30.0**	**3.10**	**55.6**	40.0	**41.2**
	RGB-expanded	0.0	1.10	11.1	**100.0**	**41.2**
Sol	Baseline	0.0	**1.40**	**0.0**	30.0	19.0
	RGB-expanded	**10.0**	0.50	**0.0**	**90.0**	**23.5**
Fable	Baseline	0.0	0.90	–	30.0	22.2
	RGB-expanded	**40.0**	**2.90**	55.6	**100.0**	**42.5**

5 Conclusion

Embodied Agent Arena evaluates nine VLM agents' native task decisions across five capabilities under source-specific interfaces. Astra has the lowest pose errors and highest full-set planning success; Opus 5.5 leads affordance overlap, contact validity, and manipulation success. The strongest manipulation result still leaves more than half of the assigned tasks incomplete. Endpoint and final-state errors prevent completion, while repeated decisions consume time and tokens without consistently resolving the remaining conditions. Per-round RGB can also reverse control rankings in the paired experiment.

The central challenge is to turn local precision into complete, timely execution: an embodied generalist must preserve achieved conditions while satisfying the remaining goals within its interaction budget. The arena tests this ability on bounded tasks with public assets. Repeated trials, longer tasks, and unseen physical environments would test its consistency.

References

Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, et al. Do As I Can, Not As I Say: Grounding Language in Robotic Affordances. *arXiv preprint arXiv:2204.01691*, 2022.

Eduardo Arnold, Jamie Wynn, Sara Vicente, Guillermo Garcia-Hernando, Áron Monszpart, Victor Adrian Prisacariu, Daniyar Turmukhambetov, and Eric Brachmann. Map-free Visual

Relocalization: Metric Pose Relative to a Single Image. *arXiv preprint arXiv:2210.05494*, 2022.

Vineet Bhat, Sungsu Kim, Valts Blukis, Greg Heinrich, Prashanth Krishnamurthy, Ramesh Karri, Stan Birchfield, Farshad Khorrami, et al. BOP-ASK: Object-Interaction Reasoning for Vision-Language Models. *arXiv preprint arXiv:2511.16857*, 2025.

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0: A Vision-Language-Action Flow Model for General Robot Control. *arXiv preprint arXiv:2410.24164*, 2024.

Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S Ryoo, Grecia Salazar, Pannag R Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong Tran, Vincent Vanhoucke, Steve Vega, Quan H Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: Robotics Transformer for Real-World Control at Scale. In *Proceedings of Robotics: Science and Systems*, 2023. doi: 10.15607/RSS.2023.XIX.025.

Nicolas Carion, Laura Gustafson, Yuan-Ting Hu, Shoubhik Debnath, Ronghang Hu, Didac Suris, Chaitanya Ryali, Kalyan Vasudev Alwala, Haitham Khedr, Andrew Huang, Jie Lei, Tengyu Ma, Baishan Guo, Arpit Kalla, Markus Marks, Joseph Greer, Meng Wang, Peize Sun, Roman Rädle, Triantafyllos Afouras, Effrosyni Mavroudi, Katherine Xu, Tsung-Han Wu, Yu Zhou, Liliane Momeni, Rishi Hazra, Shuangrui Ding, Sagar Vaze, Francois Porcher, Feng Li, Siyuan Li, Aishwarya Kamath, Ho Kei Cheng, Piotr Dollár, Nikhila Ravi, Kate Saenko, Pengchuan Zhang, and Christoph Feichtenhofer. SAM 3: Segment anything with concepts. *arXiv preprint arXiv:2511.16719*, 2025.

Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, et al. RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation. *arXiv preprint arXiv:2506.18088*, 2025a.

Zeren Chen, Xiaoya Lu, Zhijie Zheng, Pengrui Li, Lehan He, Yijin Zhou, Jing Shao, Bohan Zhuang, and Lu Sheng. Geometrically-constrained agent for spatial reasoning. *arXiv preprint arXiv:2511.22659*, 2025b. doi: 10.48550/arXiv.2511.22659.

Dongzhou Cheng, Taoran Yi, Ye Fang, Xingwu Zhang, Fan Feng, Yixuan Li, Gengxiong Zhuang, Rongze Wang, Shuai Yang, Wei Song, Weizhi Xue, Minyan Wu, Jie Gui, Jiaqi Wang, and Tong Wu. In-Context Robot Learning with VLM Agents. *arXiv preprint arXiv:2609.19138*, 2026.

Zhili Cheng, Yuge Tu, Ran Li, Shiqi Dai, Jinyi Hu, Shengding Hu, Jiahao Li, Yang Shi, et al. EmbodiedEval: Evaluate Multimodal LLMs as Embodied Agents. *arXiv preprint arXiv:2501.11858*, 2025.

Seokju Cho, Ryo Hachiuma, Abhishek Badki, Hang Su, Byung-Kwan Lee, Chan Hee Song, Sifei Liu, Subhashree Radhakrishnan, Seungryong Kim, Yu-Chiang Frank Wang, and Min-Hung Chen. SpatialClaw: Rethinking action interface for agentic spatial reasoning. Preprint, 2026.

Daniel DeTone, Tomasz Malisiewicz, and Andrew Rabinovich. SuperPoint: Self-supervised interest point detection and description. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2018.

Junjia Du, Yadi Liu, Hongcheng Guo, Jiawei Wang, Haojian Huang, Yunyi Ni, and Zhoujun Li. DependEval: Benchmarking LLMs for Repository Dependency Understanding. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 7150–7179. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-acl.373.

Letian Fu, Justin Yu, Karim El-Refai, Ethan Kou, Haoru Xue, Huang Huang, Wenli Xiao, Guanzhi Wang, et al. CaP-X: A Framework for Benchmarking and Improving Coding Agents for Robot Manipulation. *arXiv preprint arXiv:2603.22435*, 2026.

Dibya Ghosh, Homer Rich Walke, Karl Pertsch, Kevin Black, Oier Mees, Sudeep Dasari, Joey Hejna, Tobias Kreiman, Charles Xu, Jianlan Luo, You Liang Tan, Lawrence Yunliang Chen, Quan Vuong, Ted Xiao, Pannag R Sanketi, Dorsa Sadigh, Chelsea Finn, and Sergey Levine. Octo: An Open-Source Generalist Robot Policy. In *Proceedings of Robotics: Science and Systems*, 2024. doi: 10.15607/RSS.2024.XX.090.

Pengyuan Guo, Zhonghao Mai, Zhengtong Xu, Kaidi Zhang, Quan Khanh Luu, Heng Zhang, Zichen Miao, Arash Ajoudani, Zachary Kingston, Qiang Qiu, and Yu She. PLanAR: Planning-Language-Grounded Agentic Reasoning for Robot Manipulation. *arXiv preprint arXiv:2602.01662*, 2026.

Yanjiang Guo, Yen-Jen Wang, Lihan Zha, and Jianyu Chen. DoReMi: Grounding Language Model by Detecting and Recovering from Plan-Execution Misalignment. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

Haojian Huang, Zexi Li, Junhao Guo, Yehang Zhang, Wenxuan Peng, Bohan Zhou, Weilin Ruan, Leyi Wu, Chenxu Wang, Jianchong Su, Binghui Xie, Wosong Chen, Yingjie Xu, Tianhao Zhou, Suzeyu Chen, Pukun Zhao, Jiaqi He, Xinyi Li, Runze Li, Peiran Dong, Shaoxiang Dang, Jing Huang, Yingbing Chen, Yifan Chang, Tianyi Zhang, Shiyuan Deng, Haozhi Wang, Yangkai Wei, Wenqian Li, Han Yang, Kaiwen Zhou, Huaping Liu, James Cheng, Rui Shao, Donglin Wang, Yaochu Jin, Jianye Hao, Ying-Cong Chen, and Yinchuan Li. In-context learning for robots: Methods and applications. *arXiv preprint arXiv:2609.36012*, 2026a.

Haojian Huang, Chuanyu Qin, Yinchuan Li, and Yingcong Chen. Find, Fix, Reason: Context Repair for Video Reasoning. In *Proceedings of the International Conference on Machine Learning*, 2026b.

Haojian Huang, Jiahao Shi, Yinchuan Li, and Yingcong Chen. Affordance agent harness: Verification-gated skill orchestration. *arXiv preprint arXiv:2605.00663*, 2026c.

Wenlong Huang, Chen Wang, Ruohan Zhang, Yunzhu Li, Jiajun Wu, and Li Fei-Fei. VoxPoser: Composable 3D Value Maps for Robotic Manipulation with Language Models. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 540–562. PMLR, 2023a.

Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner Monologue: Embodied Reasoning through Planning with Language Models. In *Proceedings of The 6th Conference on Robot Learning*, volume 205 of *Proceedings of Machine Learning Research*, pp. 1769–1782. PMLR, 2023b.

Stephen James, Zicong Ma, David Rovick Arrojo, and Andrew J. Davison. RLBench: The Robot Learning Benchmark & Learning Environment. *arXiv preprint arXiv:1909.12271*, 2019.

Peter Jansen, Marc-Alexandre Côté, Tushar Khot, Erin Bransom, Bhavana Dalvi Mishra, Bodhisattwa Prasad Majumder, Oyvind Tafjord, and Peter Clark. DISCOVERYWORLD: A Virtual Environment for Developing and Evaluating Automated Scientific Discovery Agents. *arXiv preprint arXiv:2406.06769*, 2024.

Yunfan Jiang, Agrim Gupta, Zichen Zhang, Guanzhi Wang, Yongqiang Dou, Yanjun Chen, Li Fei-Fei, Anima Anandkumar, Yuke Zhu, and Linxi Fan. VIMA: Robot Manipulation with Multimodal Prompts. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 14975–15022. PMLR, 2023.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Foster, et al. OpenVLA: An Open-Source Vision-Language-Action Model. *arXiv preprint arXiv:2406.09246*, 2024.

Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabrael Levine, et al. BEHAVIOR-1K: A Human-Centered, Embodied AI Benchmark with 1,000 Everyday Activities and Realistic Simulation. *arXiv preprint arXiv:2403.09227*, 2024a.

Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen, Tony Lee, et al. Embodied Agent Interface: Benchmarking LLMs for Embodied Decision Making. *arXiv preprint arXiv:2410.07166*, 2024b.

Qiang Li, Qingsen Yan, Haojian Huang, Peng Wu, Haokui Zhang, and Yanning Zhang. Text-Visual Semantic Constrained AI-Generated Image Quality Assessment. In *Proceedings of the 33rd ACM International Conference on Multimedia*, 2025. doi: 10.1145/3746027.3755471.

Zexi Li, Yehang Zhang, Haojian Huang, Bohan Zhou, Wenqian Li, Chenxu Wang, Yifan Chang, Yangkai Wei, Tianyi Zhang, Ying-Cong Chen, Kaiwen Zhou, Yinchuan Li, and James Cheng. RACaP: Agentic Reasoning, Acting, and Coding as Policies for Evolvable Robot Learning. *arXiv preprint arXiv:2609.29394*, 2026a.

Zexi Li, Yehang Zhang, Wenqian Li, Haojian Huang, Chenxu Wang, Shiyuan Deng, Yangkai Wei, Tianyi Zhang, Binghui Xie, Bohan Zhou, Yifan Chang, Kaiwen Zhou, Ying-Cong Chen, James Cheng, and Yinchuan Li. Robo-Harness K1: Harnessing Robot-Use Agents via Perception Augmentation. *arXiv preprint arXiv:2609.29389*, 2026b.

Erich Liang, Roma Bhattacharjee, Sreemanti Dey, Rafael Moschopoulos, Caitlin Wang, Michel Liao, Grace Tan, Andrew Wang, et al. InFlux: A Benchmark for Self-Calibration of Dynamic Intrinsics of Video Cameras. *arXiv preprint arXiv:2510.23589*, 2025.

Jacky Liang, Wenlong Huang, Fei Xia, Peng Xu, Karol Hausman, Brian Ichter, Pete Florence, and Andy Zeng. Code as Policies: Language Model Programs for Embodied Control. *arXiv preprint arXiv:2209.07753*, 2022.

Philipp Lindenberger, Paul-Edouard Sarlin, and Marc Pollefeys. LightGlue: Local feature matching at light speed. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023.

Wufei Ma, Haoyu Chen, Guofeng Zhang, Yu-Cheng Chou, Jieneng Chen, Celso M de Melo, and Alan Yuille. 3DSRBench: A Comprehensive 3D Spatial Reasoning Benchmark. *arXiv preprint arXiv:2412.07825*, 2024.

Arjun Majumdar, Anurag Ajay, Xiaohan Zhang, Pranav Putta, Sriram Yenamandra, Mikael Henaff, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets, Sergio Arnaud, Karmesh Yadav, Qiyang Li, Ben Newman, Mohit Sharma, Vincent Berges, Shiqi Zhang, Pulkit Agrawal, Yonatan Bisk, Dhruv Batra, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Alexander Sax, and Aravind Rajeswaran. OpenEQA: Embodied Question Answering in the Era of Foundation Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 16488–16498, 2024.

Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *arXiv preprint arXiv:2112.03227*, 2021.

Austin Myers, Ching L. Teo, Cornelia Fermüller, and Yiannis Aloimonos. Affordance Detection of Tool Parts from Geometric Features. In *IEEE International Conference on Robotics and Automation*, 2015.

Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-Scale Simulation of Everyday Tasks for Generalist Robots. *arXiv preprint arXiv:2406.02523*, 2024.

Fei Ni, Min Zhang, Pengyi Li, Yifu Yuan, Lingfeng Zhang, Yuecheng Liu, Peilong Han, Longxin Kou, et al. Embodied Arena: A Comprehensive, Unified, and Evolving Evaluation Platform for Embodied AI. *arXiv preprint arXiv:2509.15273*, 2025.

Xavier Puig, Kevin Ra, Marko Boben, Jiaman Li, Tingwu Wang, Sanja Fidler, and Antonio Torralba. VirtualHome: Simulating Household Activities via Programs. *arXiv preprint arXiv:1806.07011*, 2018.

Yu Qi, Haibo Zhao, Ziyu Guo, Siyuan Ma, Ziyan Chen, Yaokun Han, Renrui Zhang, Zitiantao Lin, et al. Dissecting Embodied Abilities in Multimodal Language Models through Skill-level Evaluation and Diagnosis. *arXiv preprint arXiv:2510.08759*, 2025.

Shengyi Qian and David F. Fouhey. Understanding 3D Object Interaction from a Single Image. *arXiv preprint arXiv:2305.09664*, 2023.

Hanoona Rasheed, Mohammed Irfan Kurpath, Bin Ren, Hisham Cholakkal, Fahad Shahbaz Khan, and Salman Khan. Hard vision, easy vision: What GPT-6 Astra reveals across computer vision. *arXiv preprint arXiv:2609.35718*, 2026.

Pierre Sermanet, Tianli Ding, Jeffrey Zhao, Fei Xia, Debidatta Dwibedi, Keerthana Gopalakrishnan, Christine Chan, Gabriel Dulac-Arnold, Sharath Maddineni, Nikhil J Joshi, Pete Florence, Wei Han, Robert Baruch, Yao Lu, Suvir Mirchandani, Peng Xu, Pannag Sanketi, Karol Hausman, Izhak Shafran, Brian Ichter, and Yuan Cao. RoboVQA: Multimodal Long-Horizon Reasoning for Robotics. *arXiv preprint arXiv:2311.00899*, 2023.

Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks. *arXiv preprint arXiv:1912.01734*, 2019.

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. *arXiv preprint arXiv:2010.03768*, 2020.

Mohit Shridhar, Lucas Manuelli, and Dieter Fox. CLIPort: What and Where Pathways for Robotic Manipulation. *arXiv preprint arXiv:2109.12098*, 2021.

Ruiyang Si, Jianxin Bi, Shunyu Yang, Rui Ni, Wenbo Huang, Qiang Wang, Shulong Jiang, Duomin Wang, Xiuyu Li, Haiwen Feng, Zhen Dong, and Daquan Zhou. Fewer Tokens, Better Action: GPT-6 Astra Robot Agents with 14% Higher Success Rate but 65% Fewer Tokens. *arXiv preprint arXiv:2610.01939*, 2026.

Ishika Singh, Valts Blukis, Arsalan Mousavian, Ankit Goyal, Danfei Xu, Jonathan Tremblay, Dieter Fox, Jesse Thomason, and Animesh Garg. ProgPrompt: Generating Situated Robot Task Plans using Large Language Models. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 11523–11530, 2023. doi: 10.1109/ICRA48891.2023.10161317.

Chan Hee Song, Valts Blukis, Jonathan Tremblay, Stephen Tyree, Yu Su, and Stan Birchfield. RoboSpatial: Teaching Spatial Understanding to 2D and 3D Vision-Language Models for Robotics. *arXiv preprint arXiv:2411.16537*, 2024.

Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, et al. ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI. *arXiv preprint arXiv:2410.00425*, 2024.

Alexander Veicht, Paul-Edouard Sarlin, Philipp Lindenberger, and Marc Pollefeys. GeoCalib: Single-image calibration with geometric optimization. In *European Conference on Computer Vision*, 2024.

Hanqing Wang, Shaoyang Wang, Yiming Zhong, Zemin Yang, Jiamin Wang, Zhiqing Cui, Jiahao Yuan, Yifan Han, Mingyu Liu, and Yuexin Ma. Affordance-R1: Reinforcement Learning for Generalizable Affordance Reasoning in Multimodal Large Language Model. *arXiv preprint arXiv:2508.06206*, 2025a.

Qineng Wang, Baiqiao Yin, Pingyue Zhang, Jianshu Zhang, Kangrui Wang, Zihan Wang, Jieyu Zhang, Keshigeyan Chandrasegaran, et al. MindCube: Spatial Mental Modeling from Limited Views. *arXiv preprint arXiv:2506.21458*, 2025b.

Ruoyao Wang, Peter Jansen, Marc-Alexandre Côté, and Prithviraj Ammanabrolu. ScienceWorld: Is your Agent Smarter than a 5th Grader? *arXiv preprint arXiv:2203.07540*, 2022.

Dongming Wu, Yanping Fu, Saike Huang, Yingfei Liu, Fan Jia, Nian Liu, Feng Dai, Tiancai Wang, Rao Muhammad Anwer, Fahad Shahbaz Khan, and Jianbing Shen. Rag-net: Large-scale reasoning-based affordance segmentation benchmark towards general grasping. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11980–11990, October 2025.

Leyi Wu, Yifan Zhao, Jinjie Zhang, Suzeyu Chen, Wosong Chen, Zhifei Chen, Tianshuo Xu, Qingchun He, Hongxin Hu, Haojian Huang, Yangkai Wei, Wenqian Li, Yinchuan Li, and Ying-Cong Chen. RoboStressBench: Benchmarking VLM Robustness to Physical Visual Stress in Embodied Scenes. *arXiv preprint arXiv:2606.00828*, 2026.

Runsen Xu, Weiyao Wang, Hao Tang, Xingyu Chen, Xiaodong Wang, Fu-Jen Chu, Matt Feiszli, and Kevin J. Liang. Multi-SpatialMLLM: Multi-Frame Spatial Understanding with Multi-Modal Large Language Models. *arXiv preprint arXiv:2505.17015*, 2025.

Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in Space: How Multimodal Large Language Models See, Remember, and Recall Spaces. *arXiv preprint arXiv:2412.14171*, 2024.

Rui Yang, Hanyang Chen, Junyu Zhang, Mark Zhao, Cheng Qian, Kangrui Wang, Qineng Wang, Teja Venkat Koripella, et al. EmbodiedBench: Comprehensive Benchmarking Multi-modal Large Language Models for Vision-Driven Embodied Agents. *arXiv preprint arXiv:2502.09560*, 2025a.

Sihan Yang, Runsen Xu, Yiman Xie, Sizhe Yang, Mo Li, Jingli Lin, Chenming Zhu, Xiaochen Chen, et al. MMSI-Bench: A Benchmark for Multi-Image Spatial Intelligence. *arXiv preprint arXiv:2505.23764*, 2025b.

Shiduo Zhang, Zhe Xu, Peiju Liu, Xiaopeng Yu, Yuan Li, Qinghui Gao, Zhaoye Fei, Zhangyue Yin, et al. VLABench: A Large-Scale Benchmark for Language-Conditioned Robotics Manipulation with Long-Horizon Reasoning Tasks. *arXiv preprint arXiv:2412.18194*, 2024.

Yehang Zhang, Haojian Huang, Yifan Chang, Jianchong Su, Bohan Zhou, Yingjie Xu, Wosong Chen, Tianhao Zhou, Chenxu Wang, Tianyi Zhang, Yangkai Wei, Wenqian Li, Shiyuan Deng, Yinchuan Li, Ying-Cong Chen, and Zexi Li. World Action Agent: Harnessing VLMs for Robot Manipulation via World Action Rehearsal. *arXiv preprint arXiv:2609.29964*, 2026a.

Yehang Zhang, Jianchong Su, Haojian Huang, Yifan Chang, Tianhao Zhou, Xinli Xu, Yingjie Xu, Yinchuan Li, Zexi Li, and Ying-Cong Chen. WorldLines: Benchmarking and Modeling Long-Horizon Stateful Embodied Agents. In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, 2026b.

Zixin Zhang, Kanghao Chen, Hanqing Wang, Hongfei Zhang, Harold Haodong Chen, Chenfei Liao, Litao Guo, and Ying-Cong Chen. A4-Agent: An agentic framework for zero-shot affordance reasoning. *arXiv preprint arXiv:2512.14442*, 2025.

Pukun Zhao, Longxiang Wang, Chen Chen, Peicheng Wang, Fanqing Zhou, Runze Li, and Haojian Huang. STAR: Mitigating Cascading Errors in Spatial Reasoning via Turn-point Alignment and Segment-level DPO. In *IEEE International Conference on Multimedia and Expo (ICME)*, 2026a.

Pukun Zhao, Longxiang Wang, Miaowei Wang, Chen Chen, Fanqing Zhou, and Haojian Huang. EvoEmpirBench: Dynamic Spatial Reasoning with Agent-ExpVer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pp. 36564–36572, 2026b. doi: 10.1609/aaai.v40i43.40979.

Enshen Zhou, Yibo Li, Jingkun An, Jiayuan Zhang, Shanyu Rong, Mengzhen Liu, Yi Han, Yuheng Ji, Huajie Tan, Jiawei He, Pengwei Wang, Zhongyuan Wang, Cheng Chi, Lu Sheng, and Shanghang Zhang. Towards spatial trace with reasoning in vision-language models for robotics. *arXiv preprint arXiv:2512.13660*, 2026.

Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control. In *Proceedings of The 7th Conference on Robot Learning*, volume 229 of *Proceedings of Machine Learning Research*, pp. 2165–2183. PMLR, 2023.

Appendix: Supporting Evidence and Protocols

Reading guide. Appendices A–B define task construction, interfaces, and scoring. Appendix C expands Table 2, including source-composition sensitivity and dispersion for Opus 5.5 and DeepSeek-1017. Appendix D follows the five domains of Section 4.3, from full task comparisons to Manipulation's auxiliary metric definitions. Appendix E specifies the execution-protocol and multi-round review comparisons, and Appendix F illustrates recorded actions and terminal outcomes, with additional examples across all five capabilities in Appendix H. Appendix G gives the evaluation scales and public resources behind Table 1.

A Benchmark Construction and Source Inventory

Embodied Agent Arena contains 1,000 cases from 32 established sources and our GeoProbe benchmark (Table A1). A case is one query or episode; multiple queries can share a scene. Benchmark-specific allocation uses seed 42 and targets diagnostic coverage of the five capabilities in Section 3.1.

Table A1. The unified 1,000-case arena, drawn from 32 established sources and our GeoProbe benchmark. Detailed source allocations follow below.

Domain	Cases	Share (%)	Task coverage
Geometry	370	37.0	Calibration, pose, scale, tracing
Spatial Reasoning	220	22.0	Relations, views, video
Affordance	70	7.0	Functional parts, contact
Task Planning	157	15.7	Household, navigation, science
Manipulation	183	18.3	Robot operations, sequences
Total	1,000	100.0	660 fixed-input; 340 interactive

Query vocabulary. Figure 2a counts each term once per query, balances weights across source groups and domains, and includes English translations of GeoProbe queries. GeoProbe's ten protocols form one source group.

Geometry (370 cases). InFlux-Real (Liang et al., 2025) contributes 30 single-frame intrinsic-estimation cases, of which four have invalid reference annotations and are excluded from continuous-error calculations. Map-free (Arnold et al., 2022) contributes 30 relative-pose image pairs with known camera intrinsics. MultiSPA-derived uses Map-free images and poses with the camera-motion question formulation associated with Multi-SpatialMLLM (Xu et al., 2025): 12 angle, ten distance, and eight displacement-vector questions. This local adaptation retains the Map-free image and pose references. The pose transformation and camera-center displacement use distinct coordinate conventions and reference targets.

GeoProbe comprises ten diagnostic protocols: 168 queries over 130 unique RGB images, predominantly rendered in Blender, with four real-photo displacement cases. Shared scenes separate calibration, pose, depth, displacement, and scale requirements (Table A2).

TraceSpatial-Bench (Zhou et al., 2026), introduced with RoboTracer, adds paired 2D and 3D queries on 56 scenes. These 112 Geometry cases predict offline object trajectories, scored for reference agreement and geometric constraints.

Table A2. GeoProbe's ten diagnostic protocols, with shared images and scenes.

Diagnostic	Cases	Diagnostic	Cases
Object displacement, fixed camera	12	Full camera intrinsics	24
Object displacement, small view change	12	Focal length, centered principal point	12
Object displacement, large view change	12	Relative camera pose	24
Real tabletop displacement	4	Sparse depth	12
Object dimensions	12	Reference-based scale	44

Spatial Reasoning (220 cases). 3DSRBench (Ma et al., 2024) contributes 50 relation questions; 28 flip-tagged entries were displayed unflipped. Scores use corrected references for five affected answers under the unflipped presentation. MindCube (Wang et al., 2025b) contributes 50 multi-view cases: 39 among-view, six around-view, and five rotation. VSI-Bench (Yang et al., 2024) contributes 50 video questions: nine absolute-distance, nine object-size, fourteen relative-direction, six relative-distance, five counting, four room-size, two appearance-order, and one route-planning. Eight frames form its initial observation. RoboSpatial-Home (Song et al., 2024) contributes ten cases each for compatibility, configuration, and context. BOP-ASK (Bhat et al., 2025) contributes twelve relative-position and eight relative-depth questions. MMSI-Bench (Yang et al., 2025b) contributes 20 cross-image questions. Together these comprise 183 categorical, 27 numerical, and ten point-set questions, scored as defined in Appendix B.3.

Affordance (70 cases). UMD (Myers et al., 2015) contributes 30 views spanning five tool classes and six functions: grasp, cut, scoop, contain, pound, and support. Multiple views can depict the same object. The 20 ReasonAff-style cases adapt instruction-conditioned queries motivated by ReasonAff (Wang et al., 2025a) to 3DOI (Qian & Fouhey, 2023) interaction annotations. Another 20 RAGNet-3DOI cases use explicit action-conditioned queries (Wu et al., 2025). The two subsets use disjoint images. These images and annotations support native point and region evaluation.

Task Planning (157 cases). ALFRED (Shridhar et al., 2019) and ALFWorld (Shridhar et al., 2020) contribute 20 visual cases each, covering placement, heating, cooling, cleaning, containers, multiple objects, and illumination. ScienceWorld (Wang et al., 2022) contributes 73 text-based cases across eighteen topics. VirtualHome (Puig et al., 2018) contributes 21 symbolic programs in seven scenes. The composition analysis removes these 94 cases together. DiscoveryWorld (Jansen et al., 2024) contributes ten tasks with structured observations. HumanCLAW contributes thirteen Habitat navigation and sitting episodes: five sofa, three toilet, two chair, two plant, and one television target. Observations are visual, textual, or symbolic.

Table A3. Manipulation tasks across 13 sources, with identical assignments for all agents.

Benchmark	Cases	Task coverage
CALVIN (Mees et al., 2021)	10	Ten sequences, each with five ordered subgoals
CLIPort (Shridhar et al., 2021)	18	Eighteen task and color-generalization configurations
RLBench (James et al., 2019)	10	Ten task configurations
RoboCasa (Nasiriany et al., 2024)	10	Seven atomic and three composite tasks
RoboCasa365	10	Three atomic and seven composite tasks
RoboWits	10	Ten official task configurations
RoboTwin 2.0 (Chen et al., 2025a)	50	Fifty distinct manipulation tasks
VIMA-Bench (Jiang et al., 2023)	10	Six placement-generalization and four new-task cases
VLABench (Zhang et al., 2024)	25	8 ID, 3 category, 4 commonsense, 7 semantic, 3 texture
ManiSkill (Tao et al., 2024)	10	Nine binary tabletop tasks; one quadruped reward task
LIBERO-PRO (Fu et al., 2026)	10	Ten object, spatial, goal, and multi-object configurations
robosuite	5	Two lift seeds; stacking, restacking, and wiping
BEHAVIOR-1K (Li et al., 2024a)	5	Five multi-object household activities

Manipulation (183 cases). Table A3 lists 182 binary tasks and one continuous-reward task. LIBERO-PRO, robosuite, and BEHAVIOR-1K use benchmark-specific code interfaces (Fu et al., 2026). RoboCasa365 uses variations designated for pretraining in the source inventory. Appendix B.1 specifies the native assistance available to agents.

B Evaluation Protocol and Reproducibility

B.1 Observations and Model-Directed Interaction

Geometry, Spatial Reasoning, and Affordance inspect fixed observations; Planning and Manipulation interact with an environment. All domains except Manipulation use a persistent Python runtime; Manipulation uses its native agent loop. Both enforce episode-level budgets and termination. Variables and environment state persist between code rounds, and images produced by a cell reach the next model request after that cell ends.

Table A4. Model-visible observations, actions, and feedback. Private scoring annotations are unavailable to the agent.

Domain	Observation support	Actions and feedback
Geometry	Image or image pair; intrinsics for Map-free	One final numerical submission
Spatial Reasoning	Images, crops, and video-frame queries	Evidence inspection and answer submission
Affordance	Image and interaction instruction	Point and box submission; polygons in the added experiment
Task Planning	Images, text, or symbolic state	Navigation, interaction, programs, public progress
Manipulation	Protocol-specific scene/state observations and native grounding helpers	Motion or skill calls; execution and task checks

ALFRED and visual ALFWorld expose observation, search, navigation, interaction, and memory primitives. CLIPort and VIMA expose instance inspection and pick-and-place actions.

RoboTwin provides actor localization, pose inspection, planning probes, and execution. LIBERO-PRO, robosuite, and BEHAVIOR-1K expose benchmark-specific observation and action interfaces; RoboCasa includes fixture and contact localization. The model makes decisions using these native helpers; the baseline adds no specialist perception or verification pipeline. Geometry requests direct visual estimates and excludes pose-solver implementation.

A runtime exception can be followed by another code round if budget remains; prior state changes are not automatically undone. Model requests, code rounds, primitive calls, and environment steps count different units. Stored visualization frames are also distinct from images actually sent to the model. Qualitative figures sample the recorded camera streams.

B.2 Models and Execution Budgets

Table A5. Models and configured identifiers for all nine agents in the five domains.

Model label	Identifier
GPT-6 Astra	`gpt-6-astra`
GPT-6 Sol	`gpt-6-sol`
Claude Fable 5.1	`claude-fable-5-1`
Claude Opus 5.5	`claude-opus-5-5`
DeepSeek-V4.1-Flash-1017	`deepseek-flash`
Gemini 3.8 Flash	`gemini-3.8-flash`
Qwen 3.8 Max	`qwen3.8-max`
Qwen 3.5 397B-A17B	`qwen3.5-397b-a17b`
Qwen 3.5 27B	`qwen3.5-27b`

DeepSeek-1017 (DS-1017 in compact figures and tables) denotes the 17 October 2026 release DeepSeek-V4.1-Flash-1017, served by `deepseek-flash`; this alias is not an immutable model snapshot. Its evaluation was recorded on 18–19 October 2026. DeepSeek-1017 Spatial runs use no YOLOE, and 3DSRBench inputs use the corrected image flips and matching native labels. The original fixed-input and Planning configurations use temperature zero. Opus 5.5 uses the recorded identifier above and the same Baseline Manipulation protocol as the original seven agents. Baseline request limits are 1,800 output tokens and 360 seconds; fixed-input profiles also allow up to 16,384 tokens and 900 seconds. On the configured CLI interface, reasoning effort is maximum for Sol and high for Astra and Fable; API requests omit this parameter. Astra's first repetition on the original 90-case Geometry core uses the high-effort CLI configuration. Manipulation's request settings and shared execution limits are specified in Appendix E.1.

Table A6. Baseline episode budgets. Tokens accumulate within each case; rounds are code interactions, not repetitions. Dashes denote unspecified fields. Appendix B.4 lists fixed-input settings.

Entry	Code rounds	Tokens	Env. steps	Primitives	Episode seconds	Task seconds
Geometry	12	120k	–	–	–	960
Spatial: non-VSI	6	96k	–	–	900	960
Spatial: VSI	8	240k	–	–	1,200	1,500
ALFRED, ALFWorld	8	320k	120	540	1,200	1,500
DiscoveryWorld	8	320k	100	400	1,200	1,500
HumanCLAW	101	2,000k	100	400	10,800	11,000
ScienceWorld	8	320k	100	500	1,200	1,500
VirtualHome	8	320k	16	120	1,200	1,500
Affordance	12	120k	–	–	–	960

Execution and native-call time limits are clipped by remaining budgets. Appendix E.1 gives Manipulation's shared execution limits, camera presets, and native step-limit semantics.

B.3 Scoring and Repeated Evaluation

Submission and termination. A final submission commits a fixed-input prediction. Interactive runs end when the agent stops, the environment reaches a terminal condition, or a source-specific round, token, action, or time limit is reached. The evaluator scores the submitted output or recorded state independently of the agent's declaration. A complete episode includes all its internal reasoning, actions, and recovery; these steps are not separate attempts.

Task scores and aggregation. Categorical questions use answer accuracy and interactive tasks use native goal success. Invalid or absent predictions remain in success denominators. Invalid reference annotations are excluded before scoring. Continuous errors pool valid predictions with compatible metrics and units, alongside their coverage. Numerical Spatial Reasoning uses exact-match success, $|\hat{y} - y| < 10^{-6}$, alongside continuous errors; this differs from VSI-Bench's official mean relative accuracy. Point-set questions retain their local set criterion.

Pass@1 and repeated scores. For N assigned cases and R repetitions, let $s_{ir} \in \{0, 1\}$ denote success on case i in run r. The average single-episode success rate is

$$\text{Pass@1} = \frac{1}{R} \sum_{r=1}^{R} \left(\frac{1}{N} \sum_{i=1}^{N} s_{ir} \right). \tag{2}$$

This estimates success in one complete episode under the specified condition. Fresh agent state is used for formal repetitions. For any per-run score M_r, we compute

$$\bar{M} = \frac{1}{R} \sum_{r=1}^{R} M_r, \qquad v_M = \frac{1}{R-1} \sum_{r=1}^{R} (M_r - \bar{M})^2, \qquad \text{SD}_M = \sqrt{v_M}. \tag{3}$$

Tables using $\bar{M} \pm \text{SD}_M$ report variation between run-level scores. Variance is supplied separately in squared metric units. Figure 4 averages valid-case errors within each run and then weights the recorded runs equally. Its coverage sensitivity instead fixes the cases valid in every shared repetition for each model pair; those common-cohort estimates are reported separately. Figure 5a–b uses per-run categorical accuracies; panel c uses full-source native Pass@1. The original pairwise gaps average accuracy differences over shared run indices,

with the comparator fixed per subtask. Run indices align reporting repetitions; seeds and tool configurations can differ. Run-to-run SD requires $R > 1$.

Scalar estimates and depth. For J valid targets, $e_j = \hat{y}_j - y_j$ gives $\mathrm{MAE} = J^{-1}\sum_j |e_j|$ and $\mathrm{RMSE} = \sqrt{J^{-1}\sum_j e_j^2}$. Absolute relative error is $\mathrm{AbsRel} = J^{-1}\sum_j |e_j|/|y_j|$ on nonzero targets, and $\mathrm{MAPE} = 100\,\mathrm{AbsRel}$ is expressed in percent. Size and focal-length estimates use absolute and relative errors. Depth additionally reports RMSE in meters. Depth errors are computed over positive finite predictions within each image, then averaged across images, with valid-point coverage reported. The fraction within 10% relative error and $\delta_1 = J_{\mathrm{all}}^{-1}\sum_j \mathbf{1}[\max(\hat{y}_j/y_j, y_j/\hat{y}_j) < 1.25]$ use all queried points as their denominator; invalid predictions contribute zero. Ordinal accuracy is the fraction of eligible point pairs with the correct depth ordering, with invalid predictions counted as incorrect. Scale alignment removes the global depth factor.

Displacement and camera pose. For a nonzero scalar or vector target y, motion relative error is $\|\hat{y} - y\|_2/\|y\|_2$. The MultiSPA-derived tolerance is 20%; a zero target requires zero absolute error. GeoProbe object-displacement diagnostics separately test movement status, direction, absolute distance error, and direction agreement with distance error at most 1 cm. Stationary cases are evaluated without division by zero. Camera translation error is $e_t = \|\hat{\mathbf{t}} - \mathbf{t}\|_2$, reported in centimeters. Rotation error is $e_R = \arccos[(\mathrm{tr}(R^{\mathsf{T}}\hat{R}) - 1)/2]$, in degrees, clipping the argument to $[-1, 1]$. GeoProbe reports component errors and joint success within 2° and 2 cm.

Camera calibration. Focal MAPE averages the absolute relative errors of f_x and f_y per case. Principal-point error is $e_c = \|\hat{\mathbf{c}} - \mathbf{c}\|_2$ in pixels, also normalized by the image diagonal D. Joint calibration requires both focal components to satisfy 10% relative error and $e_c \le 0.02D$. Main-table pose errors and complementary continuous metrics use pooled valid-observation means; Figure 4 instead uses equal-run means, as specified above.

Spatial tracing. For each case, predicted and reference paths are independently resampled to $L = 100$ equally spaced arc-length positions. Trajectory error is $\mathrm{RMSE}_{\mathrm{trace}} = \sqrt{L^{-1}\sum_{\ell=1}^{L}\|\hat{\mathbf{p}}_\ell - \mathbf{p}_\ell\|_2^2}$. We average this per-case value over valid scenes. The 2D metric uses normalized image coordinates; the 3D metric uses meters. The two metrics retain their respective units. The supplied variance is the population variance of the 56 per-scene RMSE values for a single run.

Offline 3D success requires valid initiation, target arrival, and collision checks to pass together. The predicted start must lie within 20 cm of the object point cloud or the reference start. At least one of the final three predicted points must lie within 20 cm of the target 3D box; for two-point paths, only the endpoint is checked. Translating the object point cloud to each checked predicted position must yield at most 20% overlap with the environment. Collision is checked at the sampled positions. Missing or invalid trajectories count as failures. 2D tracing reports continuous error. Qwen-27B trajectory outputs undergo format normalization that preserves all predicted coordinate values.

Native affordance grounding. For a directly predicted bounding box B and ground-truth mask G, the box interior is rasterized and $\mathrm{IoU}_{B,G} = |B \cap G|/|B \cup G|$. Contact validity is $\mathbf{1}[p \in G]$ for the submitted point p, with a reference-box fallback only where a mask is unavailable. A separate diagnostic uses $\mathrm{IoU}_{\mathrm{box}} = |B \cap B_G|/|B \cup B_G|$ against the reference box B_G. Strict box success requires $\mathrm{IoU}_{\mathrm{box}} \ge 0.5$; contact validity is scored independently. The source field named `giou` contains ordinary box–mask IoU, not generalized IoU. The separate polygon protocol rasterizes the directly submitted polygons into a region P and scores $|P \cap G|/|P \cup G|$. It adds no external segmenter or iterative segmentation refinement; Appendix C.5 gives output validation and coverage.

Resource accounting. Inference tokens, action calls, model rounds, and elapsed time measure different aspects of execution. The Manipulation comparison reports token usage and action calls from the supplied execution analysis; protocol-matched per-case durations are unavailable. Request, trial, and native-call time limits are specified separately in Appendix E.1.

B.4 Repetition Protocol and Result Accounting

Repetitions. In the original seven-agent cohort, non-tracing Geometry, Spatial Reasoning, and Affordance use five repetitions per case for six models and three for Fable. Opus 5.5 uses five runs for non-tracing Geometry and one for tracing, Spatial Reasoning, Affordance, Planning, and Manipulation. DeepSeek-1017 uses five runs for non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance. Tracing, Planning, Manipulation, and direct-polygon prediction use one run per assignment for all models. Run-to-run SD is reported only for repeated measurements. Manipulation's observation schedules appear in Appendices E.1 and C.4.

Additional execution configurations. Fixed-input evaluation also uses per-case limits of up to 48 code rounds and 480k tokens for Geometry and Affordance, 24 rounds and 384k tokens for non-VSI Spatial Reasoning, and 32 rounds and 960k tokens for VSI. Task time limits reach 7,200 seconds, or 10,800 seconds for VSI. Budgets vary by source and execution configuration.

Result selection. In the original cohort, fixed-input and Planning trial slots use the first valid outcome, independent of correctness. Manipulation uses the fixed adopted baseline and RGB-expanded manifests. The original Baseline CSV covers 182 binary assignments. Table 2 preserves the original eight models and adds DeepSeek-1017, whose RGB-expanded Manipulation setting uses the limits in Table A16. The Opus 5.5 Manipulation score uses the same Baseline source interfaces and is included in the main-table comparison. Appendix E.1 defines missing-record accounting, shared-current sources, and the common subset used in Table 3.

Coverage and scoring denominators. Table A7 reports the original cohort and DeepSeek-1017 coverage. Opus 5.5 metric-specific valid counts are in Table A12; no whole-domain valid count is inferred from overlapping summary metrics. Opus 5.5 HumanCLAW uses the final completed record for each of the 13 assignments: navigation succeeds at 20 cm proximity, while sitting requires the native interaction condition. Two RLBench exceptions and one RoboTwin exception count as unsuccessful assignments; Opus 5.5 Manipulation remains 87/182 (47.8%). InFlux excludes four invalid references. Binary criteria retain all assigned tasks. Reported success requires the native evaluator, rather than an agent declaration. Superseded HumanCLAW attempts are not counted as additional assignments. Manipulation covers 182 binary tasks, including all five BEHAVIOR-1K assignments per model; AnymalC-Spin is excluded from this binary comparison. Continuous metrics retain finite valid predictions.

Table A7. Original seven agents and DeepSeek-1017 (DS-1017). Valid submission coverage for fixed-input domains and recorded terminal outcomes for Planning. Assigned denominators include repetitions and invalid-reference slots; invalid references are excluded from metric scoring. Manipulation uses the fixed allocation in Table A3.

Domain	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Geometry	1382/1402	1382/1402	819/886	1382/1402	1314/1402	1382/1402	1376/1402	1356/1402
Spatial Reasoning	1060/1100	1100/1100	586/660	1097/1100	1077/1100	1033/1100	982/1100	954/1100
Affordance	350/350	350/350	208/210	350/350	335/350	349/350	349/350	348/350
Task Planning	157/157	157/157	157/157	157/157	157/157	157/157	157/157	157/157

Manipulation diagnostics. CALVIN reports average consecutive goals completed. Other auxiliary metrics average task-normalized native criteria or control evidence over fixed applicable subsets. They differ in their scoring items and evidence time windows; Appendix D.5 defines the aggregation, coverage, and missing-record treatment.

Main-table metric scope. Geometry rotation and translation pool the 30 Map-free and 24 GeoProbe pose cases in degrees and centimeters. Spatial Pass@1 covers all 220 cases; numerical AbsRel covers 27, and categorical accuracy separately covers 183. Original Affordance box–mask IoU and contact validity cover all three sources, with 70 cases. Polygon IoU evaluates the same allocation under the direct-polygon protocol. Planning uses terminal success across 157 cases for every model. Manipulation uses task-weighted native success across 182 binary assignments; Tables A9 and A11 report each source separately.

Dispersion and valid subsets. Pooled continuous statistics use $\bar{x} = K^{-1}\sum_{k=1}^{K} x_k$ and $v_{\text{pool}} = K^{-1}\sum_k (x_k - \bar{x})^2$ over valid case–run observations. This dispersion differs from the variance of run means in Equation 3. Main-table continuous metrics use pooled means; the task-level Geometry and Spatial analyses use equal-run means and SD, with common-valid-cohort checks in Appendices D.1 and D.2.

C Complete Model Comparisons

This section supports Section 4.2 at three resolutions: complementary native metrics, differences among source benchmarks, and the metric profiles shown in Figure 3. Task-level explanations follow in Appendix D.

C.1 Metrics and Source-Level Results

Table A8 gives complementary native metrics. Table A9 adds DeepSeek-1017 to the original-cohort source results, and Table A11 adds Opus 5.5, with denominators and repetition rules from Appendix B.4. **Reading the metric families.** GeoProbe focal MAPE combines full-intrinsics and focal-only queries; displacement MAE combines its four camera/environment conditions. Scale MAPE combines object dimensions and numerical reference-scale queries, while qualitative scale judgments retain Pass@1. TraceSpatial reports 2D and 3D errors separately because their coordinates and units differ. Each metric retains the target's physical interpretation.

Table A8. Complementary native metrics across four domains; DS-1017 abbreviates DeepSeek-V4.1-Flash-1017. Repeated binary rates show the mean above ± run SD; single-run rates have no SD. Continuous values are pooled valid-observation means; Table A12 distinguishes case–run variance from variance of run means. Polygon IoU uses a separate single run for all nine models. Planning uses all 157 cases for every model. N counts eligible cases. GP denotes GeoProbe; Manipulation source results appear in Tables A9 and A11.

Metric	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	Opus 5.5	DS-1017
Geometry (370 cases)										
InFlux focal MAPE (%)↓	26	53.02	47.79	52.65	55.50	52.36	67.82	65.86	60.27	62.91
InFlux principal error (px)↓	26	3.82	3.82	3.82	3.82	3.88	3.82	3.82	3.82	3.57
Angle relative L2↓	12	0.760	0.915	1.32	1.41	0.962	1.30	2.04	1.166	1.504
Distance relative L2↓	10	0.294	0.418	0.785	0.560	0.419	0.411	0.331	0.455	0.479
Vector relative L2↓	8	0.425	0.591	0.625	0.454	0.647	1.02	1.03	0.516	0.750
Map-free rotation (°)↓	30	29.94	46.60	52.75	35.11	46.32	63.03	64.94	40.35	52.92
Map-free translation (cm)↓	30	188.21	265.85	259.75	243.06	280.99	430.82	374.96	240.27	325.64

Complementary metrics, continued.

Metric	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	Opus 5.5	DS-1017
Map-free direction (°)↓	30	26.28	48.91	53.02	38.92	52.42	95.15	78.78	43.56	64.88
GP focal MAPE (%)↓	36	6.92	9.57	31.71	14.67	26.59	33.39	38.56	16.48	33.99
GP principal error (px)↓	24	51.95	110.62	63.40	78.41	66.83	58.23	59.05	60.46	75.40
Depth MAE (m)↓	12	0.307	0.633	0.412	0.490	0.589	0.934	0.944	0.510	0.688
Depth RMSE (m)↓	12	0.338	0.753	0.517	0.598	0.699	1.05	1.06	0.599	0.814
Depth AbsRel↓	12	0.104	0.229	0.140	0.163	0.193	0.312	0.292	0.169	0.229
Displacement MAE (cm)↓	40	0.585	0.973	1.49	1.52	2.28	2.48	2.94	1.164	1.934
GP rotation (°)↓	24	1.17	2.54	2.83	2.21	3.10	7.55	5.26	3.05	3.75
GP translation (cm)↓	24	5.36	9.50	9.30	6.99	7.85	23.59	12.05	11.13	14.54
Scale MAPE (%)↓	44	1.90	2.84	3.76	3.77	3.93	16.07	16.45	3.91	7.62
Scale judgment Pass@1 (%)↑	12	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	100.0 ±0.0	93.3 ±7.0	100.0 ±0.0	98.3 ±3.7
2D trace RMSE↓	56	0.120	0.132	0.118	0.118	0.127	0.155	0.284	0.180	0.141
3D trace RMSE (m)↓	56	0.382	0.463	0.435	0.469	0.426	0.629	0.682	0.683	0.470
3D constraint success (%)↑	56	33.9	21.4	25.0	17.9	37.5	8.9	7.1	8.9	14.3
Spatial Reasoning (220 cases)										
Native Pass@1 (%)↑	220	69.4 ±1.0	60.7 ±2.5	62.0 ±4.6	68.9 ±1.1	65.4 ±1.3	50.6 ±1.8	48.0 ±0.9	58.6	51.6 ±1.5
Categorical Pass@1 (%)↑	183	82.2 ±0.6	71.3 ±3.0	72.7 ±5.2	81.2 ±1.3	76.7 ±1.9	60.0 ±2.2	56.8 ±0.7	69.4	61.3 ±1.7
Numerical AbsRel↓	27	0.213	0.273	0.186	0.205	0.161	0.251	0.288	0.222	0.310
Affordance (70 cases)										
BBox–GT box IoU↑	70	0.390	0.366	0.386	0.382	0.322	0.146	0.021	0.449	0.278
BBox–GT mask IoU↑	70	0.286	0.264	0.284	0.272	0.250	0.116	0.013	0.329	0.217
Polygon–GT mask IoU↑	70	0.379	0.325	0.328	0.365	0.291	0.092	0.051	0.397	0.249
BBox success@0.5 (%)↑	70	46.6 ±1.3	43.7 ±1.6	46.2 ±0.8	45.7 ±1.7	33.1 ±11.8	5.4 ±3.7	0.3 ±0.6	54.3	28.9 ±3.7
Contact-point validity (%)↑	70	60.3 ±2.6	52.0 ±3.6	57.1 ±2.5	52.9 ±4.2	44.6 ±13.0	32.6 ±4.7	6.3 ±1.6	62.9	50.9 ±4.2
Task Planning (157 cases)										
Native Pass@1 (%)↑	157	80.3	44.6	63.7	35.7	48.4	17.8	19.1	48.4	31.2

Table A9. Source results for the original seven agents and DeepSeek-1017 (DS-1017); Opus 5.5 appears in Table A11. Spatial/Affordance: mean above ± run SD (%); Planning: solved/assigned cases; Manipulation: success (%). N counts eligible cases.

Source	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Spatial Reasoning: categorical accuracy (%)									
3DSRBench	50	73.6 ±2.6	71.2 ±1.8	72.7 ±3.1	74.0 ±3.7	70.0 ±2.4	69.2 ±3.6	57.6 ±2.6	61.6 ±7.1
BOP-ASK	20	96.0 ±4.2	87.0 ±4.5	88.3 ±2.9	92.0 ±2.7	97.0 ±2.7	85.0 ±3.5	84.0 ±4.2	88.0 ±7.6
MindCube	50	90.8 ±1.1	68.0 ±7.3	79.3 ±8.1	90.8 ±2.7	89.2 ±3.0	46.0 ±2.8	50.4 ±4.1	54.4 ±4.3
MMSI-Bench	20	77.0 ±5.7	55.0 ±11.7	66.7 ±7.6	55.0 ±8.7	56.0 ±9.6	35.0 ±9.4	29.0 ±7.4	53.0 ±5.7
RoboSpatial	20	85.0 ±3.5	81.0 ±8.9	88.3 ±2.9	88.0 ±5.7	76.0 ±4.2	90.0 ±3.5	92.0 ±5.7	87.0 ±5.7
VSI-Bench	23	72.2 ±7.3	70.4 ±5.7	36.2 ±28.9	83.5 ±8.4	65.2 ±5.3	44.3 ±6.4	39.1 ±3.1	37.4 ±5.0
Affordance: strict box success at IoU ≥ 0.5 (%)									
ReasonAff-style	20	19.0 ±5.5	6.0 ±4.2	16.7 ±7.6	13.0 ±5.7	5.0 ±0.0	0.0 ±0.0	0.0 ±0.0	3.0 ±4.5
RAGNet-3DOI	20	14.0 ±4.2	15.0 ±5.0	13.3 ±2.9	7.0 ±2.7	6.0 ±2.2	4.0 ±2.2	1.0 ±2.2	5.0 ±6.1
UMD	30	86.7 ±3.3	88.0 ±4.5	87.8 ±5.1	93.3 ±2.4	70.0 ±27.0	10.0 ±7.8	0.0 ±0.0	62.0 ±6.9

Source-level comparisons, continued.

Source	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Task Planning: terminal success (single run)									
ALFRED	20	15/20	3/20	7/20	3/20	2/20	1/20	0/20	1/20
ALFWorld	20	14/20	4/20	6/20	2/20	4/20	3/20	4/20	4/20
DiscoveryWorld	10	10/10	4/10	7/10	3/10	5/10	1/10	1/10	2/10
HumanCLAW	13	3/13	2/13	1/13	0/13	3/13	0/13	0/13	2/13
ScienceWorld	73	64/73	38/73	59/73	29/73	45/73	11/73	14/73	28/73
VirtualHome	21	20/21	19/21	20/21	19/21	17/21	12/21	11/21	12/21
Total	157	126/157	70/157	100/157	56/157	76/157	28/157	30/157	49/157
Manipulation: task success (%)									
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0	10.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7	33.3
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0	20.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0	4.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0	20.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0	60.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0	36.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0	11.1
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0	50.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0	80.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Total	182	41.8	19.8	24.7†	9.3	18.7	9.9	11.0	20.9

†See Appendix E.1 for coverage and aggregation.

C.2 Sensitivity to Source Composition

Table A10. Original seven agents and DeepSeek-1017. Task-composition analysis (success %). Spatial uses only the 183 categorical queries; Planning removes ScienceWorld and VirtualHome; Affordance removes UMD and uses strict box success. All columns retain the assigned-case denominator of the indicated subset.

Agent	Spatial without VSI	Planning without text/symbolic	Affordance without UMD
Astra	83.6	66.7	16.5
Sol	71.4	20.6	10.5
Fable	77.9	33.3	15.0
Qwen-Max	80.9	12.7	10.0
Gemini	78.4	22.2	5.5
Qwen-397B	62.2	7.9	2.0
Qwen-27B	59.4	7.9	0.5
DeepSeek-1017	64.8	14.3	4.0

Removing ScienceWorld and VirtualHome changes Astra's Planning success from 80.3% to 66.7%, and Qwen-Max's from 35.7% to 12.7%. The same table isolates categorical Spatial Reasoning without VSI-Bench and strict affordance-box success without UMD. These subset comparisons measure sensitivity to the task mixture.

Equal source weighting. For the original seven-agent cohort, to test whether source allocation drives the lead, we average complete source scores with equal weight, $M_{\text{source}} = B^{-1} \sum_{b=1}^{B} M_b$,

retaining every assigned case within each source. Across all six Spatial sources, Astra scores 71.9% versus Qwen-Max's 69.1%; across all three Affordance sources, contact validity is 54.2% versus Fable's 50.9%; across all six Planning sources, success is 75.2% versus Fable's 53.1%. These are the highest non-Astra macro scores within the original seven-agent cohort. The ranking therefore persists under both case weighting and equal source weighting. These analyses use the complete domain allocations, separate from the 46-case multi-round review diagnostic in Section 4.4.

C.3 Opus 5.5 Repetitions, Dispersion, and Coverage

Opus 5.5 was assigned five runs on the original 258 Geometry cases; tracing and all other domains have one run. Four invalid InFlux reference cases are excluded from eligible scoring. Continuous means and variances pool valid case–run observations. Population variance uses divisor K and squared metric units; it is not a standard error or a run-to-run SD. Table A12 retains the effective sample counts. For example, pooled pose rotation is 23.3°, while the equally weighted mean of five run means is 23.4°; differing valid counts explain the distinction.

Additional Opus 5.5 results and scope. Planning includes all 157 assignments. The five non-HumanCLAW sources contribute 73 successes, and HumanCLAW adds 3/13, for 76/157 (48.4%). On HumanCLAW, Opus 5.5 succeeds on 3/5 navigation tasks and 0/8 sitting tasks. It finds the target and reaches within 20 cm on 7/13 cases, including 4/8 sitting tasks; proximity does not establish interaction success. These are local fixed-subset results, not official leaderboard scores. Opus 5.5 Manipulation uses the same Baseline protocol as the original main table; its 87/182 score counts two RLBench exceptions and one RoboTwin exception as unsuccessful assignments. The original paired-protocol analysis is retained separately. The supplemental recordings in Appendix H illustrate individual episodes; they do not establish the per-round image-input condition used in the paired protocol comparison.

Execution diagnostics added to the main figures. Opus 5.5 completes at least one through five consecutive CALVIN goals on 80%, 70%, 40%, 40%, and 10% of the ten assignments. Summing these survival rates gives 2.40 consecutive goals on average, as plotted in Figure 8b. On the five ManiSkill diagnostic tasks, it satisfies all final goal conditions on PickCube, PlaceSphere, and StackCube, and none on PegInsertionSide or PlugCharger; their mean condition score is 60% (Figure 8c). These are Baseline observations, with no corresponding Opus 5.5 RGB-expanded run. Opus 5.5 3D tracing has 8.9% joint success across 56 assignments, alongside the 0.683 m RMSE in Table A12.

Table A11. Additional Opus 5.5 source results. Planning includes all six sources. Manipulation retains all binary assignments, including execution errors. Spatial values cover all questions in each source, not only categorical questions.

Domain	Source	Assigned	Success (%)
Spatial	3DSRBench	50	60.0
Spatial	BOP-ASK	20	95.0
Spatial	MindCube	50	74.0
Spatial	MMSI-Bench	20	45.0
Spatial	RoboSpatial-Home	30	53.3
Spatial	VSI-Bench	50	36.0

Domain	Source	Assigned	Success (%)
Planning	ALFRED	20	15.0
Planning	ALFWorld	20	45.0
Planning	DiscoveryWorld	10	30.0
Planning	ScienceWorld	73	58.9
Planning	VirtualHome	21	71.4
Planning	HumanCLAW	13	23.1
Planning	All assigned	157	48.4
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	61.1
Manipulation	RLBench	10	30.0
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	10.0
Manipulation	RoboWits	10	10.0
Manipulation	RoboTwin 2.0	50	46.0
Manipulation	VIMA-Bench	10	100.0
Manipulation	VLABench	25	76.0
Manipulation	ManiSkill	9	55.6
Manipulation	LIBERO-PRO	10	80.0
Manipulation	Robosuite	5	100.0
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	All assigned	182	47.8

Table A12. Opus 5.5 continuous statistics. Variance is over valid case–run observations (population divisor). Coverage is valid/eligible; it includes repetitions.

Metric	Mean	Variance	Coverage
InFlux focal MAPE (%)	60.27	641	130/130
InFlux principal error (px)	3.82	27.7	130/130
Angle relative L2	1.166	3.59	60/60
Distance relative L2	0.455	0.787	50/50
Vector relative L2	0.516	0.267	40/40
Map-free rotation (°)	40.35	1.52e+03	142/150
Map-free translation (cm)	240.27	2.99e+04	142/150
Map-free direction (°)	43.56	1.67e+03	142/150
GP focal MAPE (%)	16.48	341	180/180
GP principal error (px)	60.46	2.35e+03	120/120
Depth MAE (m)	0.510	0.167	60/60
Depth RMSE (m)	0.599	0.179	60/60
Depth AbsRel	0.169	0.0167	60/60
Displacement MAE (cm)	1.164	2.2	200/200
GP rotation (°)	3.05	26.4	120/120
GP translation (cm)	11.13	378	120/120
Scale MAPE (%)	3.91	48.9	220/220
2D trace RMSE	0.180	0.0207	56/56
3D trace RMSE (m)	0.683	0.147	56/56
Numerical AbsRel	0.222	0.052	27/27
BBox–GT box IoU	0.449	0.151	70/70
BBox–GT mask IoU	0.329	0.0897	70/70
ReasonAff / BBox IoU	0.240	0.131	20/20
ReasonAff / Box–mask IoU	0.203	0.104	20/20
RAGNet-3DOI / BBox IoU	0.222	0.133	20/20

Metric	Mean	Variance	Coverage
RAGNet-3DOI / Box–mask IoU	0.172	0.0784	20/20
UMD / BBox IoU	0.740	0.0271	30/30
UMD / Box–mask IoU	0.517	0.0252	30/30

Table A13. Opus 5.5 Geometry variation across five run means. Variance is the sample variance of the five run means (divisor four), rather than pooled case–run variance.

Metric	Mean of runs	Run variance	Valid counts (R1–R5)
Pose rotation (°)	23.4	13.62	47/54/54/53/54
Pose translation (cm)	135.8	162.3	47/54/54/53/54
InFlux focal MAPE (%)	60.27	2.232	26/26/26/26/26
InFlux principal error (px)	3.82	0	26/26/26/26/26
Angle relative L2	1.166	0.01284	12/12/12/12/12
Distance relative L2	0.455	0.004787	10/10/10/10/10
Vector relative L2	0.516	0.001763	8/8/8/8/8
Map-free rotation (°)	40.95	65.58	23/30/30/29/30
Map-free translation (cm)	243.1	1283	23/30/30/29/30
Map-free direction (°)	44.29	82.1	23/30/30/29/30
GP focal MAPE (%)	16.48	37.63	36/36/36/36/36
GP principal error (px)	60.46	17.03	24/24/24/24/24
Depth MAE (m)	0.51	0.08685	12/12/12/12/12
Depth RMSE (m)	0.5988	0.09093	12/12/12/12/12
Depth AbsRel	0.1695	0.01156	12/12/12/12/12
Displacement MAE (cm)	1.164	0.2254	40/40/40/40/40
GP rotation (°)	3.048	0.9234	24/24/24/24/24
GP translation (cm)	11.13	2.178	24/24/24/24/24
Scale MAPE (%)	3.909	7.49	44/44/44/44/44

C.4 DeepSeek-1017: Protocol, Dispersion, and Task Outcomes

DeepSeek-1017 denotes DeepSeek-V4.1-Flash-1017, the 17 October 2026 release served by `deepseek-flash`. Its main evaluation contains 3,192 assigned case–run slots covering all 1,000 cases; the added polygon experiment contributes 70 separate slots. The supplied archive fixes one adopted record per task and run; superseded transport and adapter attempts remain provenance records rather than additional trials. Non-tracing Geometry, Spatial Reasoning, and point-and-box Affordance have five runs; tracing, Planning, and Manipulation have one. Four invalid InFlux references are excluded from eligible scoring. Failed or absent binary predictions retain their assigned denominator; continuous errors use finite valid predictions. Five-run SDs measure variation between run means.

Observations and execution budgets. Spatial runs use no YOLOE; 3DSRBench's corrected flipped inputs use matching native labels. In Manipulation, both Baseline and RGB-expanded use native RGB observations and tools. RGB-expanded supplies current views at every decision, whereas Baseline follows each source's native observation schedule. DeepSeek-1017 uses the former and completes 38/182 tasks. Its source-specific round, token, and in-episode time ceilings (Table A16) match those in Table A34. The paired observation experiment covers the original seven agents. Original Affordance uses native points and rectangles. A separate

direct-polygon run scores all 70 assigned cases, yielding 0.249 IoU; its 67 valid outputs and three zero-scored failures are detailed in Appendix C.5.

Local evidence and complete task outcomes. The source breakdown in Table A15 connects aggregate performance to task demands. DeepSeek-1017 succeeds on 2/5 HumanCLAW navigation tasks and 0/8 sitting tasks. It solves 49/157 Planning assignments, falling to 9/63 when text/symbolic ScienceWorld and VirtualHome are removed. CALVIN averages 1.40 consecutive goals, while only 1/10 full five-goal sequences succeeds. In the five-task ManiSkill diagnostic, none of the final condition sets is fully satisfied. BEHAVIOR-1K has no complete success in five assignments: pre-termination logs confirm localization in four and a grasp candidate in three. Only stages confirmed before termination enter the diagnostic.

Table A14. DeepSeek-1017 pooled statistics and metric-specific coverage. Variance is across valid case–run predictions for continuous metrics; rate variances are across run scores (population divisor). Continuous coverage counts finite valid predictions; rate coverage retains failed or absent submissions as zero. Polygon variance is across all 70 assigned cases, including three zero-scored failures. Single-run results have no repeated-run uncertainty estimate.

Metric	Mean	Variance	Scored/eligible
Pose rotation (°)	30.325	1.72e+03	259/270
Pose translation (cm)	182.702	6.44e+04	259/270
Spatial Pass@1 (%)	51.636	1.79	1100/1100
Numerical AbsRel	0.310	0.0704	120/135
Box–mask IoU	0.217	0.0549	348/350
Contact validity (%)	50.857	14.4	350/350
Polygon–mask IoU	0.249	0.0725	70/70
Planning success (%)	31.210	—	157/157
Manipulation success (%)	20.879	—	182/182

Table A15. DeepSeek-1017 native source outcomes. Spatial includes all question types; Affordance uses the original point-or-box criterion. Manipulation measures complete-task success.

Domain	Source	Cases	Success (%)
Spatial	3DSRBench	50	61.6 ± 7.1
Spatial	BOP-ASK	20	88.0 ± 7.6
Spatial	MMSI-Bench	20	53.0 ± 5.7
Spatial	MindCube	50	54.4 ± 4.3
Spatial	RoboSpatial-Home	30	58.0 ± 3.8
Spatial	VSI-Bench	50	20.0 ± 2.0
Affordance	RAGNet-3DOI	20	39.0 ± 7.4
Affordance	ReasonAff-style	20	15.0 ± 3.5
Affordance	UMD	30	86.0 ± 6.4
Planning	ALFRED	20	5.0
Planning	ALFWorld	20	20.0
Planning	DiscoveryWorld	10	20.0
Planning	HumanCLAW	13	15.4
Planning	ScienceWorld	73	38.4
Planning	VirtualHome	21	57.1
Manipulation	BEHAVIOR-1K	5	0.0
Manipulation	CALVIN	10	10.0
Manipulation	CLIPort	18	33.3

Domain	Source	Cases	Success (%)
Manipulation	LIBERO-PRO	10	50.0
Manipulation	ManiSkill	9	11.1
Manipulation	RLBench	10	20.0
Manipulation	RoboCasa	10	0.0
Manipulation	RoboCasa365	10	0.0
Manipulation	RoboTwin2	50	4.0
Manipulation	RoboWits	10	20.0
Manipulation	Robosuite	5	80.0
Manipulation	VIMA-Bench	10	60.0
Manipulation	VLABench	25	36.0

Table A16. DeepSeek-1017 Manipulation ceilings. Tokens are cumulative; minutes are in-episode wall-clock limits. Process wrappers allow additional startup and teardown time. Each assigned task is one episode. ManiSkill success counts nine binary tasks; its additional locomotion case has a continuous reward.

Source	Model rounds	Tokens	Minutes
CALVIN	40	1280k	120
BEHAVIOR-1K	24	640k	180
LIBERO-PRO	24	640k	180
Robosuite	24	640k	180
CLIPort	24	960k	60
ManiSkill	32	640k	60
RLBench	16	640k	60
RoboCasa	24	1120k	60
RoboCasa365	24	1120k	60
RoboTwin2	16	640k	60
RoboWits	24	640k	60
VIMA-Bench	16	640k	60
VLABench	16	640k	60

Table A17. DeepSeek-1017 execution stages. Fractions count confirmed events per assignment, with zero credit for unverified events. BEHAVIOR-1K uses pre-termination evidence.

Source	Diagnostic	Confirmed / assigned
CALVIN	All five ordered goals completed	1/10
ManiSkill	All final conditions on the five-task subset	0/5
RoboCasa	Approach; grasp and lift; stable release	3/4; 3/4; 0/4
RoboCasa365	Approach; grasp and lift; stable release	3/7; 3/7; 0/7
BEHAVIOR-1K	Localization; grasp candidate	4/5; 3/5
	Arrival; holding; held motion; release	0/5 each

C.5 Direct Polygon Grounding

The nine-model Affordance experiment retains the same 70 images and interaction instructions, with 20 RAGNet, 20 ReasonAff-style/3DOI, and 30 UMD cases. Each model directly predicts polygons in image coordinates alongside its point and bounding box. Rasterizing the polygons yields a predicted mask for IoU and Dice; no SAM, detector, or iterative segmentation refinement is added. This is a separate output protocol, not a re-scoring of the original rectangular

predictions. The main table preserves the original box–mask result and adds polygon IoU; the original point results remain in Table A8.

Coverage and aggregation. Table A18 reports the final evaluation results for the single-run polygon protocol. Invalid polygons and absent final submissions score zero. The original point/box pass flag is not treated as polygon success. No run-to-run SD is inferred from one run. DeepSeek-1017 yields 67 valid polygons: two outputs have vertices outside the image, and one case reaches the 12-round limit without a submission. All three contribute zero to the 70-case mean (24.92%). Its existing thinking-enabled, high-effort setting uses a 1,800-token visible-response ceiling; the 120k-token case budget also counts reasoning. The other limits remain 12 rounds and 900 seconds per case. Each of the 70 cases contributes one adopted attempt.

Table A18. Polygon–mask IoU (%), with one run on 20 RAGNet, 20 ReasonAff-style, and 30 UMD cases.

Model	RAGNet	ReasonAff	UMD	Overall
Astra	19.19	10.60	68.52	37.88
Sol	13.22	7.40	62.17	32.54
Fable	12.95	17.00	56.51	32.77
Qwen-Max	6.26	12.75	63.28	36.47
Gemini	10.92	4.21	57.71	29.06
Qwen-397B	5.86	2.45	16.02	9.24
Qwen-27B	3.98	0.30	9.05	5.10
Opus 5.5	18.55	13.29	71.49	39.74
DeepSeek-1017	8.67	9.22	46.21	24.92

Opus 5.5 has the highest aggregate at 39.74%, followed by Astra at 37.88%, while both score far lower on the interaction-conditioned scenes than on UMD tools (Figure 6b).

C.6 Interpreting the Capability Profiles

Figure 3 retains the original fixed visualization scales. Geometry is $50[\exp(-r/45)+\exp(-t/150)]$, where r and t are the pooled rotation error in degrees and translation error in centimeters. Spatial is $[p + 100\exp(-a/0.25)]/2$, where p is native Pass@1 (%) and a is numerical AbsRel. Affordance is $[100u + c]/2$, where u is original box–mask IoU and c is contact validity (%). Planning and Manipulation use their task success rates (%). G, S, A, P, and M denote these five domains. These monotone scales summarize complementary metrics; they are not native task tolerances or success probabilities.

The five-domain index is the equally weighted mean of the five indices. Opus 5.5 includes all 157 Planning assignments, scoring 76/157 (48.4%), and all 182 Baseline Manipulation assignments. The Affordance index uses box–mask IoU and contact validity from Table A8; polygon IoU is reported as an additional metric.

DeepSeek-1017 uses the same transformations, giving G/S/A/P indices of 40.3/40.3/36.3/31.2 and a Manipulation score of 20.9. Its Manipulation axis marks the per-round RGB schedule. Panel b averages the eight Baseline agents; panels a and c show all nine profiles.

D Task-Level Analysis of Astra

The subsections follow Section 4.3 and Figures 4–8. The tables add DeepSeek-1017 wherever matching task-level records are available. Original seven-agent peer-selection and paired-protocol diagnostics remain explicitly scoped; Opus 5.5 source results appear in Appendix C.3. Each first identifies the task requirement behind the main-text claim, then gives the supporting comparison and broader task coverage. Repetition and valid-coverage conventions are defined in Appendix B.4.

D.1 Geometry: Task Requirements and Source Conditions

Task-level comparisons. Table A19a supplies the absolute errors behind Figure 4a. Its relative reduction is $100(1 - e_{\text{Astra}}/e_{\text{peer}})$, using the lowest-error non-Astra peer in the original cohort for each target. Equal-run means retain task-specific units: MAPE for focal length, size, and height, and $100 \times \text{AbsRel}$ for depth. Astra leads the five GeoProbe targets; Sol leads on InFlux focal estimation.

Coverage sensitivity. Table A19b restricts each Astra–peer comparison to cases valid in every shared run. All six comparison directions remain unchanged. The ranking is thus stable on the common valid cohort.

Table A19. Geometric errors (%, lower is better), mean ± run SD. (a) Original seven agents plus DeepSeek-1017 (DS-1017); GP denotes GeoProbe. (b) Common-valid cohorts with matched cases and runs.

(a) All valid cases

Task	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
GP focal-only	6.2±2.1	11.2±5.5	29.5±1.8	13.8±3.1	23.7±12.9	38.6±3.2	38.4±5.0	38.1±10.5
InFlux focal	53.0±2.1	47.8±2.0	52.6±2.2	55.5±1.0	52.4±3.4	67.8±1.8	65.9±2.1	62.9±1.6
GP depth	10.4±2.9	22.9±2.7	14.2±2.7	16.3±1.3	19.2±2.1	31.2±2.8	29.2±1.3	22.9±2.5
GP object size	6.3±1.0	8.5±0.7	7.6±0.2	9.6±0.8	8.3±0.3	11.4±0.6	9.4±0.3	8.9±0.7
GP ref. height	0.31±0.11	0.90±0.36	2.31±0.08	1.69±0.72	2.73±0.75	17.55±1.22	13.69±0.86	5.71±1.49
GP height ratio	0.32±0.15	0.95±0.41	2.23±0.17	2.17±0.22	2.67±0.28	20.35±0.87	17.08±2.70	8.35±5.02

(b) Matched common-valid cohorts

Task	Cases	Rounds	Peer	Astra	Peer
GP focal-only	12	5	Sol	6.21±2.10	11.16±5.47
InFlux focal	26	5	Sol	53.02±2.15	47.79±2.00
GP depth	9	3	Fable	10.16±3.49	14.22±2.88
GP object size	12	3	Fable	6.44±1.36	7.63±0.18
GP ref. height	12	5	Sol	0.31±0.11	0.90±0.36
GP height ratio	12	5	Sol	0.32±0.15	0.95±0.41

Camera motion and real photographs. The rendered displacement subsets share 12 object interventions while changing the camera condition. Table A20 compares Astra with Sol, the lowest-MAE non-Astra agent under each displacement condition. The fixed-camera advantage persists when the two stationary Blender cases are excluded. Under the native direction-and-1 cm criterion, both agents achieve 96.7% fixed-camera success: the continuous error distinguishes precision that this tolerance does not resolve.

The four real-photo pairs form a separate tabletop subset. Their displacement labels are supplied annotations, without independent camera or metric calibration. The Blender comparison uses different scenes and object motions.

Table A20. Displacement MAE (cm), mean ± run SD. Sol is the strongest original non-Astra comparator; DeepSeek-1017 is added under the same conditions. Moving-only excludes stationary cases.

Condition	Cases	Astra	Sol	DeepSeek-1017
Blender: fixed camera	12	0.240±0.109	0.467±0.267	1.517±0.105
Blender: fixed, moving-only	10	0.288±0.131	0.560±0.321	1.780±0.076
Blender: small view change	12	0.567±0.158	0.678±0.152	2.022±0.346
Blender: large view change	12	1.125±0.144	1.958±0.295	2.492±0.908
Real photographs	4	0.050±0.068	0.415±0.236	1.250±0.234

Coupled motion and metric scale reveal distinct estimation errors. Table A21 separates the GeoProbe pose modes: mixed rotation and translation produce 14.16 cm translation error and zero joint success, despite much lower errors on simpler motions. The best non-Astra translation estimate on the mixed subset is Qwen-Max at 16.77 cm; no evaluated agent achieves joint success on this subset. Translation-only cases have zero target rotation. Depth also separates ordering from metric scale: Astra's ordinal accuracy is 96.3%, whereas raw AbsRel is 0.104; oracle scale alignment reduces it to 0.038. Alignment uses reference depth.

Explicit scale references and regular rendered geometry support these estimates.

Table A21. GeoProbe camera-pose modes. Errors are pooled valid-observation means; joint success uses 2°/2 cm tolerances and shows mean ± run SD.

Motion	Model	Cases	Rotation (°)	Translation (cm)	Joint success (%)
Stationary	Astra	4	0.00	0.00	100.0±0.0
	DeepSeek-1017	4	0.00	0.00	100.0±0.0
Rotation only	Astra	6	0.35	1.00	96.7±7.5
	DeepSeek-1017	6	5.43	13.82	13.3±13.9
Translation only	Astra	6	0.00	1.55	76.7±25.3
	DeepSeek-1017	6	1.18	11.03	20.0±13.9
Rotation + translation	Astra	8	3.26	14.16	0.0±0.0
	DeepSeek-1017	7	6.35	25.24	0.0±0.0

TraceSpatial-Bench: path matching and boundary constraints. Astra has lower 3D path RMSE than Gemini (0.382 versus 0.426 m), but fewer jointly valid paths (19/56 versus 21/56; Table A8). Among Astra's trajectories, 27 fail initiation, 24 fail arrival, and ten fail collision checks; these sets overlap. Of its 37 joint failures, 36 violate initiation or arrival, while only one satisfies both boundaries and fails collision avoidance alone. This supports the boundary-anchoring diagnosis in Figure 4c. The offline tolerances and sampled-position collision rule are defined in Appendix B.3.

D.2 Spatial Reasoning: Cross-Source Task Contrasts

Complete task coverage. Table A23 covers all 43 groups across six sources. Source categories are retained; MindCube uses task axes and question templates, prioritizing composed rotation and turn–translation before individual motion and relation queries.

Figure labels and comparisons. Figure 5a shows BOP-ASK left/right (L/R), MindCube relations (Rel.) and camera translation (Shift), 3DSRBench actor-relative side (Actor) and object orientation (Facing), and VSI hard directions (Hard). Panel b compares BOP camera depth (Near/Far), 3DSR camera/object distance (Cam./Obj.), and VSI relative distance (Rel.); each displayed peer leads or ties on a row. Source scenes are retained; the Obj. row has two cases.

Shared-round gaps. The original diagnostic below selects one task per source and its highest-mean original non-Astra peer; Figure 5c now shows native source scores. Gaps and SD then use shared run indices (Table A22; protocol in Appendix B.3). On object-facing queries, Fable's three-run mean is 72.2% versus Astra's 50.0%; Astra averages 56.7% across all five runs.

Table A22. Original source-specific Astra–peer comparisons. Accuracy and gaps use shared available runs and are in percentages and percentage points. Run-index alignment follows Appendix B.3.

Task	Peer	Rounds	Astra	Peer	Gap ± SD
MMSI camera–region	Fable	3	100.0	77.8	+22.2±19.2
VSI medium direction	Qwen-Max	5	100.0	83.3	+16.7±11.8
BOP-ASK left/right	Gemini	5	95.0	96.7	−1.7±7.0
MindCube camera shift	Qwen-Max	5	78.5	89.2	−10.8±14.0
3DSR object facing	Fable	3	50.0	72.2	−22.2±9.6
RoboSpatial configuration	Q-397B	5	78.0	100.0	−22.0±4.5

Numerical error and acceptance are complementary. The numerical rows in Table A23 use the exact-match criterion from Appendix B.3. Table A24a instead measures continuous error on valid predictions. Fixing Astra's cohort across runs preserves low object-size AbsRel (0.091), while absolute-distance coverage is only 25/45 submissions (Table A24b). The table reports error together with valid-prediction coverage.

Table A23. All Spatial task groups, with DeepSeek-1017 (DS-1017) added to the original cohort. Entries are mean ± run SD (%); bold means mark the highest score in each row, including ties. Astra's column is shaded. N counts distinct cases; each assigned case remains in the success denominator.

Task	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
3DSRBench									50 cases
World height	4	60.0±13.7	50.0±0.0	**66.7**±14.4	55.0±11.2	25.0±0.0	50.0±0.0	50.0±0.0	25.0±0.0
Above / below	7	**77.1**±7.8	60.0±12.0	71.4±0.0	**77.1**±12.8	74.3±12.0	74.3±6.4	68.6±12.0	60.0±21.2
Camera depth	10	90.0±7.1	98.0±4.5	93.3±5.8	90.0±0.0	92.0±4.5	96.0±5.5	**100.0**±0.0	92.0±8.4
Next to	1	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	0.0±0.0	**60.0**±54.8	0.0±0.0	0.0±0.0
Object distance	2	70.0±27.4	80.0±27.4	83.3±28.9	**100.0**±0.0	**100.0**±0.0	50.0±0.0	80.0±27.4	50.0±0.0
Parallel / perpendicular	1	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	20.0±44.7	**100.0**±0.0	80.0±44.7	40.0±54.8
Same/different facing	4	55.0±11.2	50.0±0.0	50.0±0.0	50.0±0.0	50.0±0.0	**75.0**±17.7	35.0±13.7	40.0±13.7
Object-facing side	6	56.7±9.1	53.3±13.9	**72.2**±9.6	66.7±0.0	66.7±0.0	36.7±18.3	23.3±9.1	56.7±19.0
In front of actor	4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	95.0±11.2	90.0±13.7
Actor left/right	7	**74.3**±6.4	71.4±14.3	61.9±8.2	71.4±10.1	65.7±7.8	45.7±12.0	5.7±7.8	**74.3**±6.4
Object viewpoint	4	**70.0**±11.2	**70.0**±11.2	50.0±0.0	60.0±13.7	**70.0**±11.2	**70.0**±11.2	65.0±28.5	30.0±11.2
BOP-ASK									20 cases
Closer to camera	5	96.0±8.9	96.0±8.9	86.7±11.5	**100.0**±0.0	96.0±8.9	80.0±14.1	84.0±8.9	84.0±16.7
Farther from camera	3	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	66.7±0.0	40.0±14.9	86.7±18.3
Left / right	12	95.0±4.6	80.0±4.6	86.1±4.8	86.7±4.6	**96.7**±4.6	91.7±0.0	95.0±4.6	90.0±7.0
MindCube									50 cases
Behind observer	7	97.1±6.4	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	88.6±12.0	80.0±16.3	65.7±16.3	**100.0**±0.0
Camera translation	13	78.5±10.0	64.6±8.8	56.4±16.0	**89.2**±6.9	84.6±5.4	20.0±6.9	35.4±12.9	27.7±14.0
Imagined observer	2	**90.0**±22.4	70.0±27.4	50.0±0.0	80.0±27.4	70.0±27.4	30.0±27.4	30.0±27.4	60.0±22.4
Object relations	12	**96.7**±4.6	63.3±12.6	91.7±8.3	85.0±3.7	95.0±7.5	31.7±7.0	35.0±3.7	45.0±9.5
Composed rotation	5	**100.0**±0.0	96.0±8.9	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	96.0±8.9	96.0±8.9
Turn then translate	11	**90.9**±6.4	43.6±7.6	75.8±10.5	**90.9**±6.4	87.3±10.4	49.1±13.8	58.2±10.4	47.3±4.1
MMSI									20 cases
Attribute (Appr.)	2	**90.0**±22.4	70.0±27.4	50.0±0.0	60.0±22.4	50.0±0.0	70.0±27.4	40.0±22.4	60.0±41.8
Attribute (Meas.)	2	**100.0**±0.0	50.0±0.0	66.7±57.7	40.0±22.4	50.0±0.0	0.0±0.0	40.0±22.4	80.0±27.4

Table A23 continued. Values are mean ± run SD (%).

Task	N	**Astra**	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
MSR	2	40.0±41.8	30.0±27.4	**50.0**±0.0	40.0±22.4	30.0±27.4	30.0±44.7	40.0±22.4	**50.0**±0.0
Motion (Cam.)	1	80.0±44.7	80.0±44.7	**100.0**±0.0	**100.0**±0.0	60.0±54.8	0.0±0.0	0.0±0.0	60.0±54.8
Motion (Obj.)	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	80.0±44.7	80.0±44.7	80.0±44.7	20.0±44.7	80.0±44.7
Camera–camera relation	3	**100.0**±0.0	73.3±27.9	**100.0**±0.0	66.7±0.0	73.3±14.9	26.7±14.9	33.3±0.0	80.0±18.3
Camera–object relation	1	0.0±0.0	40.0±54.8	**66.7**±57.7	0.0±0.0	60.0±54.8	40.0±54.8	0.0±0.0	20.0±44.7
Camera–region relation	3	**100.0**±0.0	60.0±14.9	77.8±19.2	66.7±0.0	73.3±14.9	66.7±23.6	53.3±18.3	46.7±18.3
Object–object relation	1	**100.0**±0.0	80.0±44.7	**100.0**±0.0	60.0±54.8	20.0±44.7	80.0±44.7	60.0±54.8	80.0±44.7
Object–region relation	4	**50.0**±0.0	30.0±11.2	25.0±25.0	45.0±20.9	**50.0**±30.6	5.0±11.2	0.0±0.0	15.0±22.4
RoboSpatial									30 cases
Placement compatibility	10	92.0±4.5	92.0±8.4	**96.7**±5.8	90.0±0.0	64.0±8.9	80.0±7.1	90.0±7.1	92.0±13.0
Configuration	10	78.0±4.5	70.0±10.0	80.0±0.0	86.0±11.4	88.0±4.5	**100.0**±0.0	94.0±8.9	82.0±8.4
Free-space point set	10	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
VSI-Bench									50 cases
Appearance order	2	40.0±22.4	80.0±27.4	50.0±50.0	**100.0**±0.0	40.0±41.8	40.0±22.4	80.0±27.4	10.0±22.4
Absolute distance	9	2.2±5.0	2.2±5.0	3.7±6.4	0.0±0.0	**6.7**±6.1	0.0±0.0	2.2±5.0	0.0±0.0
Object count	5	32.0±11.0	20.0±14.1	**40.0**±20.0	36.0±16.7	36.0±16.7	28.0±11.0	12.0±11.0	16.0±8.9
Direction: easy	4	**100.0**±0.0	85.0±22.4	33.3±28.9	**100.0**±0.0	85.0±13.7	30.0±20.9	45.0±11.2	50.0±17.7
Direction: hard	4	45.0±11.2	30.0±27.4	25.0±43.3	**65.0**±28.5	60.0±22.4	30.0±32.6	5.0±11.2	0.0±0.0
Direction: medium	6	**100.0**±0.0	73.3±9.1	38.9±38.5	83.3±11.8	66.7±11.8	40.0±14.9	20.0±13.9	43.3±14.9
Relative distance	6	50.0±20.4	**76.7**±22.4	38.9±9.6	**76.7**±9.1	56.7±14.9	60.0±19.0	53.3±13.9	46.7±21.7
Object size	9	4.4±6.1	**22.2**±7.9	11.1±0.0	13.3±5.0	11.1±7.9	2.2±5.0	8.9±5.0	6.7±6.1
Room area	4	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0	**0.0**±0.0
Route planning	1	**100.0**±0.0	**100.0**±0.0	33.3±57.7	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0	**100.0**±0.0

Table A24. VSI-Bench numerical estimation. (a) AbsRel, equal-run mean ± SD on each model's valid predictions; lower is better. (b) Astra's coverage and fixed-cohort sensitivity: common cases have valid predictions in every recorded round.

(a) Numerical error across agents

Task	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B	DS-1017
Object count	0.35±0.05	0.40±0.11	0.20±0.11	0.23±0.11	0.31±0.08	0.41±0.06	0.42±0.04	0.34±0.09
Absolute distance	0.28±0.11	0.42±0.05	0.28±0.07	0.26±0.02	0.12±0.03	0.29±0.09	0.31±0.08	0.48±0.09
Object size	0.09±0.01	0.09±0.02	0.10±0.01	0.12±0.02	0.05±0.01	0.13±0.02	0.19±0.02	0.14±0.07
Room area	0.24±0.03	0.19±0.03	0.21±0.04	0.26±0.07	0.27±0.04	0.22±0.05	0.26±0.10	0.35±0.12

(b) Astra coverage and common-cohort error

Task	Cases	Valid / assigned	Common cases	Common AbsRel
Object count	5	25/25	5	0.350±0.053
Absolute distance	9	25/45	2	0.279±0.083
Object size	9	42/45	8	0.091±0.017
Room area	4	20/20	4	0.240±0.029

Other task requirements. RoboSpatial separates placement compatibility (92.0%), configuration (78.0%), and free-space point sets (0.0%) for Astra. All agents in this table fail the point-set criterion. Compatibility and configuration judgments succeed more often than point-set predictions.

D.3 Affordance: Contact and Region Extent

From valid contact to functional-region extent. Table A25 compares Astra and DeepSeek-1017 across all fifteen action/hand groups, including zero-scoring conditions. UMD contact validity averages 96.7% for Astra and 94.4% for Fable. Cutting combines 100% valid contacts and 96% strict box success with box–mask IoU of 0.361; the box IoU is 0.694. This separates successful localization from the limitations of rectangular mask coverage. Containment adds a same-representation comparison: Astra's strict box success is 44%, versus Qwen-Max's 96% (Table A26).

Table A25. Astra and DeepSeek-1017 Affordance metrics, mean ± run SD. IoU compares the native box to the GT mask; strict box success compares it to the GT box at IoU ≥ 0.5. Point and box success are percentages.

Action / hands	Model	Cases	Box–mask IoU	Point (%)	Box (%)
RAGNet-3DOI					
free movement / two hands	Astra	2	0.046±0.063	20.0±27.4	20.0±27.4
	DeepSeek-1017	2	0.179±0.070	50.0±0.0	30.0±27.4
free / one hand	Astra	1	0.384±0.351	60.0±54.8	60.0±54.8
	DeepSeek-1017	1	0.241±0.291	40.0±54.8	20.0±44.7
free / two hands	Astra	5	0.249±0.070	60.0±14.1	36.0±16.7
	DeepSeek-1017	5	0.083±0.030	28.0±11.0	4.0±8.9
pull / one hand	Astra	12	0.008±0.002	25.0±5.9	0.0±0.0
	DeepSeek-1017	12	0.053±0.019	41.7±8.3	0.0±0.0
ReasonAff-style					
free movement / two hands	Astra	5	0.270±0.055	48.0±11.0	48.0±11.0
	DeepSeek-1017	5	0.026±0.028	0.0±0.0	0.0±0.0
free / one hand	Astra	4	0.000±0.000	0.0±0.0	0.0±0.0
	DeepSeek-1017	4	0.035±0.040	0.0±0.0	0.0±0.0
free / two hands	Astra	4	0.001±0.002	0.0±0.0	0.0±0.0
	DeepSeek-1017	4	0.004±0.008	0.0±0.0	0.0±0.0
pull / one hand	Astra	4	0.098±0.033	60.0±13.7	0.0±0.0
	DeepSeek-1017	4	0.086±0.047	55.0±11.2	0.0±0.0
push / one hand	Astra	3	0.318±0.221	46.7±29.8	46.7±29.8
	DeepSeek-1017	3	0.146±0.201	26.7±27.9	20.0±29.8
UMD					
contain	Astra	5	0.472±0.067	100.0±0.0	44.0±16.7
	DeepSeek-1017	5	0.593±0.045	96.0±8.9	84.0±16.7
cut	Astra	5	0.361±0.009	100.0±0.0	96.0±8.9
	DeepSeek-1017	5	0.223±0.019	48.0±30.3	52.0±17.9
grasp	Astra	5	0.491±0.007	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.377±0.063	92.0±11.0	64.0±29.7
pound	Astra	5	0.637±0.009	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.490±0.034	96.0±8.9	44.0±16.7
scoop	Astra	5	0.504±0.054	80.0±0.0	80.0±0.0
	DeepSeek-1017	5	0.337±0.059	76.0±16.7	52.0±17.9
support	Astra	5	0.638±0.017	100.0±0.0	100.0±0.0
	DeepSeek-1017	5	0.466±0.056	88.0±17.9	76.0±16.7

Contact, extent, and shape give different comparisons. Astra leads box–mask IoU on UMD cutting, grasping, scooping, and support, while Sol leads containment and pounding. In the interaction-conditioned sources, one-handed pulling has zero strict box success for every original model despite nonzero contact validity. Table A26 identifies the best peer separately for each reported criterion. ReasonAff and RAGNet retain their original scenes and interaction annotations.

Table A26. Metric-specific Affordance peers. Values are mean ± run SD, using the repetition counts in Appendix B.4. Peers are selected from the original cohort; DS-1017 adds DeepSeek-1017 under the same criterion.

Task / criterion	Astra	Original peer	Model	DS-1017
UMD cutting: box–mask IoU	0.361±0.009	0.341±0.022	Fable	0.223±0.019
UMD cutting: strict box (%)	96.0±8.9	100.0±0.0	Fable	52.0±17.9
UMD containment: box–mask IoU	0.472±0.067	0.673±0.069	Sol	0.593±0.045
UMD containment: strict box (%)	44.0±16.7	96.0±8.9	Qwen-Max	84.0±16.7
ReasonAff pulling: point (%)	60.0±13.7	50.0±0.0	Fable	55.0±11.2
RAGNet pulling: point (%)	25.0±5.9	31.7±7.0	Qwen-Max	41.7±8.3
ReasonAff two-hand movement: point (%)	48.0±11.0	26.7±11.5	Fable	0.0±0.0
RAGNet two-hand movement: point (%)	20.0±27.4	50.0±0.0	Sol / Fable	50.0±0.0

D.4 Task Planning: Goal Families and Terminal States

Compound goals and source-specific requirements. Table A27 covers all 56 native families across the six sources. Each row compares Astra with the highest-success peer for that family, whereas Figure 7 keeps Fable fixed for source-level and household comparisons. Heating and dual placement test state change and multiple-object goal retention; ScienceWorld distinguishes experimental topics and instrument use. This preserves the task-specific advantages and reversals behind the aggregate Planning lead.

Table A27. Task Planning by native goal family, with one episode per assigned case. Entries are solved/assigned cases. One highest-success peer from the original cohort is shown per row, with DeepSeek-1017 (DS-1017) added separately; * denotes a tie with another peer. Source-level results appear in Table A9.

Goal family	Astra	Original peer	Model	DS-1017
ALFRED				
Illumination	1/2	1/2	Fable*	0/2
Placement	4/4	2/4	Sol*	0/4
Container transport	2/2	1/2	Fable*	1/2
Cleaning	2/2	1/2	Sol*	0/2
Cooling	2/4	2/4	Fable	0/4
Heating	2/3	0/3	Sol*	0/3
Dual placement	2/3	0/3	Sol*	0/3
ALFWorld				
Illumination	3/3	2/3	Fable*	2/3
Placement	3/4	2/4	Sol*	1/4
Container transport	2/2	1/2	Gemini*	1/2
Cleaning	3/4	1/4	Sol*	0/4
Cooling	0/1	0/1	Sol*	0/1
Heating	1/1	0/1	Sol*	0/1
Dual placement	2/5	1/5	Sol*	0/5
DiscoveryWorld				
Archaeology dating	1/1	1/1	Sol*	0/1
Combinatorial chemistry	1/1	1/1	Sol*	0/1
It's not rocket science	1/1	1/1	Fable	0/1
Lost in translation	1/1	1/1	Fable	0/1
Plant nutrients	1/1	0/1	Sol*	0/1
Discovery feed	1/1	1/1	Sol*	1/1
Doors	1/1	1/1	Qwen-Max	0/1
Navigation in a house	1/1	1/1	Fable*	0/1
Pick and give	1/1	1/1	Sol*	1/1
Space sick	1/1	1/1	Fable	0/1

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Goal family	Astra	Original peer	Model	DS-1017
HumanCLAW				
Navigate to contact	3/5	3/5	Gemini	2/5
Sit on target	0/8	0/8	Sol*	0/8
ScienceWorld				
Find animal	3/3	3/3	Fable	0/3
Find living thing	2/2	2/2	Sol*	2/2
Find non living thing	3/3	3/3	Sol*	3/3
Find plant	3/3	3/3	Sol*	2/3
Grow fruit	2/3	2/3	Fable	0/3
Inclined plane determine angle	2/2	2/2	Fable	2/2
Inclined plane friction named surfaces	15/16	16/16	Fable	8/16
Inclined plane friction unnamed surfaces	1/1	1/1	Fable*	0/1
Lifespan longest lived	2/2	2/2	Sol*	2/2
Lifespan longest lived then shortest lived	1/1	1/1	Sol*	1/1
Lifespan shortest lived	1/1	1/1	Sol*	1/1
Measure melting point known substance	8/8	6/8	Gemini	5/8
Measure melting point unknown substance	1/1	1/1	Sol*	0/1
Mendelian genetics known plant	1/1	1/1	Sol*	1/1
Mendelian genetics unknown plant	3/5	1/5	Qwen-Max*	0/5
Test conductivity	6/7	6/7	Gemini	0/7
Test conductivity of unknown substances	6/8	6/8	Fable	1/8
Use thermometer	4/6	5/6	Fable*	0/6
VirtualHome				
Grab	1/1	1/1	Sol*	0/1
Greet	1/1	1/1	Sol*	1/1
Lie	1/1	1/1	Sol*	1/1
Navigation or observation	1/1	1/1	Sol*	1/1
Read	2/2	2/2	Fable*	0/2
Rinse	1/1	1/1	Sol*	0/1
Sit	3/3	3/3	Sol*	1/3
Switchoff	1/1	1/1	Sol*	1/1
Switchon	7/7	7/7	Sol*	5/7
Touch	1/1	1/1	Sol*	1/1
Type	1/1	0/1	Sol*	0/1
Wash	0/1	1/1	Sol*	1/1

Attempted actions and terminal goals. All eleven Astra household failures exhaust their assigned budget. ScienceWorld's nine failures have scored task-failure outcomes with no recorded parser-no-match, invalid-action, or code-exception events; negative score sentinels are not physical-progress measures. HumanCLAW identifies a different limit (Table A28): six Astra sitting episodes find the target, five approach within 20 cm and issue a sit action, yet none succeeds. Fable records no target-finding success on these episodes. The shared zero terminal rate conceals different progress before the required body–object condition.

D.5 Manipulation: Control and Protocol Dependence

Baseline performance and coverage. Table A29 supports Table 2 and Figure 8a with all thirteen sources and the original seven agents. DeepSeek-1017 source results and execution limits are in Appendix C.4; its setting has no matched Baseline run. The baseline aggregate combines ten-source baseline results with the three sources shared across conditions. Astra records 76 successes out of 182 assignments; Fable records 45. Appendix E.1 defines denominator and coverage.

Table A28. HumanCLAW sitting diagnostics for eight assigned cases per model. Counts pool independently recorded conditions. † denotes lower bounds because process fields are missing for two Gemini cases; dashes denote unavailable process metrics. Terminal success retains all eight assignments. Opus 5.5 uses its final single-run records; near-plus-sit action counts are not supplied. DeepSeek-1017 adds 0/8 terminal sitting success; comparable process counts are not supplied.

Model	Process coverage	Target found	Within 20 cm	Near + sit issued	Success
Astra	8/8	6	5	5	0
Sol	8/8	3	1	1	0
Fable	8/8	0	0	0	0
Qwen-Max	0/8	–	–	–	0
Gemini	6/8	3†	3†	3†	0
Q-397B	0/8	–	–	–	0
Q-27B	0/8	–	–	–	0
Opus 5.5	8/8	4	4	–	0
DeepSeek-1017	–	–	–	–	0

Consecutive goals and task-specific evidence. Table A30 reports auxiliary metrics under both evaluation conditions. CALVIN reports average consecutive goals completed. The other rows average task-normalized native criteria or control evidence, with definitions below.

Table A29. Complete Manipulation success by protocol (%). Baseline totals use 182 assignments; – marks missing records. The last three sources use the same records in both protocols. The common nine-source subset isolates sources with records under both protocols.

Benchmark	*N*	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Baseline								
CALVIN	10	30.0	0.0	0.0	0.0	0.0	0.0	0.0
CLIPort	18	61.1	27.8	33.3	22.2	27.8	22.2	16.7
RLBench	10	60.0	20.0	10.0	0.0	20.0	0.0	10.0
RoboCasa	10	10.0	0.0	0.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	38.0	10.0	14.0	4.0	8.0	6.0	8.0
RoboWits	10	10.0	10.0	10.0	10.0	20.0	0.0	10.0
VIMA-Bench	10	40.0	30.0	30.0	10.0	20.0	30.0	30.0
VLABench	25	72.0	52.0	60.0	20.0	36.0	24.0	24.0
ManiSkill	9	55.6	0.0	–	0.0	11.1	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All assigned	182	41.8	19.8	24.7	9.3	18.7	9.9	11.0
Common nine sources	153	41.2	19.0	22.2	8.5	15.7	10.5	11.8
RGB-expanded								
CALVIN	10	0.0	10.0	40.0	10.0	0.0	0.0	0.0
CLIPort	18	55.6	33.3	50.0	27.8	38.9	22.2	16.7
RLBench	10	40.0	10.0	30.0	10.0	10.0	0.0	0.0
RoboCasa	10	10.0	0.0	10.0	0.0	0.0	0.0	0.0
RoboCasa365	10	0.0	0.0	20.0	0.0	0.0	0.0	0.0
RoboTwin 2.0	50	42.0	12.0	34.0	14.0	6.0	4.0	6.0
RoboWits	10	10.0	10.0	20.0	20.0	10.0	0.0	10.0
VIMA-Bench	10	100.0	90.0	100.0	70.0	60.0	20.0	20.0
VLABench	25	64.0	48.0	68.0	32.0	52.0	24.0	28.0
ManiSkill	9	11.1	0.0	55.6	0.0	0.0	0.0	0.0
LIBERO-PRO	10	40.0	30.0	60.0	10.0	40.0	0.0	10.0
robosuite	5	80.0	80.0	100.0	60.0	100.0	40.0	20.0
BEHAVIOR-1K	5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
All assigned	182	39.6	23.6	44.5	19.2	22.0	8.8	9.9

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Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Common nine sources	153	41.2	23.5	42.5	20.3	20.3	9.2	10.5

Table A30. Manipulation auxiliary metrics under both protocols. CALVIN reports average consecutive goals completed (0–5); other rows average task-normalized criteria or control evidence (%). Definitions and time windows appear in Table A31. – denotes unavailable records; the last three rows are shared.

Benchmark	N	Astra	Sol	Fable	Q-Max	Gemini	Q-397B	Q-27B
Baseline								
CALVIN goals (0–5)	10	3.10	1.40	0.90	0.80	1.80	0.00	0.10
RoboCasa	4	–	–	–	–	–	–	–
RoboCasa365	7	–	–	–	–	–	–	–
ManiSkill conditions	5	60.00	0.00	–	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67
RGB-expanded								
CALVIN goals (0–5)	10	1.10	0.50	2.90	1.30	0.70	0.00	0.00
RoboCasa	4	25.00	33.33	83.33	50.00	50.00	16.67	25.00
RoboCasa365	7	33.33	23.81	57.14	28.57	47.62	28.57	23.81
ManiSkill conditions	5	0.00	0.00	60.00	0.00	0.00	0.00	0.00
LIBERO-PRO	10	20.00	20.00	30.00	5.00	20.00	0.00	5.00
robosuite	5	60.00	60.00	80.00	40.00	80.00	30.00	16.67
BEHAVIOR-1K control	5	13.33	16.67	30.00	16.67	6.67	6.67	6.67

Comparisons with matching records. The common nine-source subset excludes ManiSkill and the three shared-current sources, leaving 153 assignments per agent and protocol. Astra has 63 successes in both configurations, but task-specific gains and losses offset; Fable increases from 34 to 65. Appendix E.1 examines the corresponding execution evidence.

Metric definitions and coverage. The 183-case Manipulation inventory contains 182 binary tasks and one continuous-return AnymalC-Spin task. AnymalC-Spin is evaluated separately through continuous return. For each model and configuration, task success is $\mathrm{SR} = 100N^{-1}\sum_i s_i$, where $s_i = 1$ requires native acceptance. Runs without native acceptance receive zero success credit, including timeouts, execution or infrastructure failures, and missing records. Each task contributes one run to the assigned denominator.

CALVIN scores a five-goal sequence as successful only when all five goals are completed in order. Its average number of consecutive goals completed is $\bar{L} = 10^{-1}\sum_{i=1}^{10} L_i$, where $L_i \in \{0, \ldots, 5\}$ is the verified consecutive prefix. Unknown prefixes contribute zero: baseline Qwen-Max has one such record, while expanded Gemini's five available records contribute seven goals over the fixed ten-case denominator, giving 0.70.

For the other auxiliary metrics, case i has K_i scoring items and verified indicators z_{ij}. Its score is $u_i = K_i^{-1}\sum_{j=1}^{K_i} z_{ij}$; the reported mean is $100N^{-1}\sum_i u_i$. Failed or unknown items contribute zero. This weights tasks equally even when K_i differs. The diagnostics summarize recorded control events on fixed task subsets using the scoring items and evidence windows in Table A31.

Table A31. Six auxiliary metrics, averaging task-normalized evidence within each subset.

Source	N	Scoring items	Evidence window
RoboCasa	4	Approach, grasp and lift, stable release	Any verified event during execution
RoboCasa365	7	Same three transport milestones	Any verified event during execution
ManiSkill	5	Two or three native conditions per task	Final verification state
LIBERO-PRO	10	Two task-specific conditions	Final state and limited process records
robosuite	5	Two lifting, three stacking, or four wiping items	Native success and terminal reward-derived evidence
BEHAVIOR-1K	5	Six control-evidence items	Before the first episode-end signal

Transport control versus destination success. RoboCasa uses ArrangeBreadBasket, LoadDishwasher, PickPlaceToasterToCounter, and RecycleBottlesByType. RoboCasa365 uses BreadAndCheese, MakeBananaMilkshake, MicrowaveThawingFridge, MixCakeFrosting, PlaceMeatInMarinade, PlaceVegetablesEvenly, and RemoveCuttingBoardItems. Door, appliance, and navigation-only tasks are outside these subsets. Approach requires executed motion and confirmed arrival; grasp-and-lift requires both grasp and lift confirmation. Stable release requires prior grasp-and-lift of the same canonically named object, followed by confirmed release and stability. Each item scores at most once per case. These events need not cover every task object, and release does not establish the requested destination relation.

Table A32. ManiSkill's five-task diagnostic. Items are evaluated at final verification; grasp history is separate. The other four binary tasks remain in complete-task success.

Task	K_i	Scored conditions
PickCube	2	Object placed at its goal; placed and robot static. The suspended goal requires no release.
PlaceSphere	3	Sphere on its support; on support and released; both with the sphere static.
StackCube	3	Cube A on B; on B and released; both with cube A static.
PlugCharger	2	Goal distance ≤ 0.005 m; distance satisfied and angular error ≤ 0.2 rad.
PegInsertionSide	2	Peg-head $x \geq -0.015$ m in the hole frame; official insertion success.

LiftPegUpright, PullCube, PushCube, and RollBall contribute to the nine-task success rate but not this diagnostic. Dashes mark unavailable Fable baseline records in both sets.

Task-specific conditions in LIBERO-PRO and robosuite. LIBERO-PRO uses fixed init-0 instances: spatial 1/7, object 2/5, goal 1/7, libero_10 1/9, spatial_swap 0, and object_swap 4. Ordinary placement scores recorded grasp and the official target relation; stove opening scores operation of the correct control and an open stove. The two longer tasks score a cup inside a microwave followed by the door being closed, or one food item followed by both items in a basket. Missing grasp/control records score zero; for the longer tasks, necessary preceding relations are confirmed only on official success.

robosuite uses cube lifting with seeds 99/50, cube stacking and restacking with seed 14, and spill wiping with seed 48. Lifting scores recorded grasp and native success; grasp history is missing. Stacking uses the fixed local reward q: $q > 0.125$ supports a grasp/lift/stack condition, $q \geq 0.5$ supports the top cube being lifted, and native success supplies the third item. A restacking zero can mask intermediate events under the two-cube airborne rule; these items remain unknown and receive zero credit. Wiping checks 25%, 50%, 75%, and complete removal. Current records verify the first three only on native complete success, rather than interpreting total reward as a wiped fraction. Consequently, Fable's 100% complete-task success and 80% auxiliary score are compatible: missing intermediate evidence lowers the latter.

Table A33. BEHAVIOR-1K control evidence. Each item records one type of control event and scores once per case; terminal BDDL goal success is evaluated separately.

Item	Positive evidence
Localization	Finite coordinates or pose from public RGB-D localization or grasp sampling.
Candidate	A nonempty pregrasp or grasp candidate set.
Arrival	Measured arrival from approach or recovery feedback for the same arm.
Holding	An explicit in-hand check or holding confirmation inside checked placement.
Held motion	Associated measured motion of at least 5 cm while holding, or movement with a passed holding check inside checked placement.
Release	Holding, active gripper opening, then no longer holding; or confirmed release inside checked placement.

The five tasks are outfit_a_basic_toolbox, can_meat, tidying_bedroom, putting_shoes_on_rack, and chop_an_onion. Records are linked by arm, name, and temporal order; physical-instance IDs are not consistently recorded. Object identity and inside/ontop relations are checked separately by the terminal goal evaluator. Held motion uses a 5 cm threshold. Control-event records cover four tasks for each of four models and the toolbox task for each of the other three. All retain the five-task denominator. Scoring uses the evaluated episodes up to their first end signal and requires direct evidence for each item. Fable's nine verified items yield $100 \times 9/(5 \times 6) = 30\%$, versus Astra's four items and 13.33%. RoboWits reports task success.

E Execution and Multi-Round Review Comparisons

E.1 Manipulation under the RGB-Expanded Protocol

Protocol and denominators. Baseline uses Table 2's source-specific observations and native tools. RGB-expanded adds current RGB at each decision. Both conditions use matched inference budgets, interaction allowances, and native execution-step limits. The comparison covers ten suites and 162 tasks: CALVIN, CLIPort, RLBench, RoboCasa, RoboCasa365, RoboTwin 2.0, RoboWits, VIMA, VLABench, and ManiSkill. Table 3b displays three suites; Table A29 gives all results. Common excludes ManiSkill because Fable's baseline records are missing, retaining nine suites and 153 tasks. LIBERO-PRO, robosuite, and BEHAVIOR-1K reuse 20 current records in both CSVs, bringing the domain total to 182. Missing baseline Fable ManiSkill and RoboCasa/365 stage records appear as dashes; the domain success aggregate credits the missing ManiSkill records with zero successes.

Observation and interaction settings. Baseline and RGB-expanded share the per-source inference and interaction budgets in Table A34, together with the same native execution-step limits. Baseline uses each source's native observations and tools; RGB-expanded provides the current scene's RGB views at every model decision. VIMA also supplies task reference images. Text interaction history and public state/perception tools remain available. Both conditions share an agent loop in which one code response can execute several actions.

Request and token accounting. The audited 7,378 expanded Astra/Sol/Fable requests across ten ordinary groups use temperature zero, with neither explicit reasoning effort nor a separate thinking budget. Of these, 7,372 request 3,600 response tokens; six use smaller limits as the case budget runs low. The configurations use a 3,600-token response ceiling and a 12-minute request timeout. Reported reasoning usage is recorded separately and counted once in the case total, avoiding duplication when included in completion usage; absent usage details remain unknown. Verifier allowance is one more than the model-round limit.

Table A34. Shared execution ceilings for Baseline and RGB-expanded. Times are minutes; token budgets are thousands. A round is a model decision, Tools counts interface calls, and Checks counts verifier calls. Baseline limits apply to all eight agents; the paired RGB-expanded comparison covers the original seven.

Source	Rounds	Tools	Checks	Tokens	Trial	Code	Native
CALVIN	40	7,000	41	1,280	120	10	10
CLIPort	24	256	25	960	60	10	10
RLBench	16	256	17	640	60	10	10
RoboCasa	24	256	25	1,120	60	6	10
RoboCasa365	24	256	25	1,120	60	8	10
RoboTwin 2.0	16	256	17	640	60	10	20
RoboWits	24	256	25	640	60	30	20
VIMA-Bench	16	256	17	640	60	10	10
VLABench	16	256	17	640	60	10	10
ManiSkill	32	1,024	33	640	60	10	10
LIBERO-PRO	24	256	25	640	180	60	60
robosuite	24	256	25	640	180	60	60
BEHAVIOR-1K	24	256	25	640	180	60	60

Tool calls, action calls, and physical steps are distinct counters. Trial time includes model waiting, code execution, and feedback within the loop; Code and Native limit one code execution and one native call, respectively. Request timeouts do not add extra time beyond Trial. Initialization, outer watchdogs, and retry backoff have separate clocks, so Trial is not identical to total wall-clock duration.

VIMA's reference-image count varies by task. The remaining seven ManiSkill inventory cases, including AnymalC-Spin, use one base view. RoboCasa/365 views are rendered scene observations. Request images and process videos are separate records: continuous video is not enabled uniformly for VLABench, LIBERO-PRO, or robosuite, while the later BEHAVIOR revision queue records at 15 fps. Video frame rate does not define model observation frequency.

Table A35. Expanded camera presets, width × height in pixels. The preset inventory covers all 183 assigned cases, including AnymalC-Spin. Multiple views enter as separate images.

Source	Views	Resolution
CALVIN	Static; gripper	200×200; 84×84
CLIPort	Front, left, right; no Oracle camera	640×480
RLBench	Front, wrist, overhead	128×128
RoboCasa/365	Left/right agent view, eye-in-hand	512×512
RoboTwin 2.0	Head, left, right	320×240
RoboWits	Ego, left/right wrist	848×480
VIMA-Bench	Front/top scene views and task reference images	256×128
VLABench	Right, left, forward, Franka wrist	480×480
ManiSkill	Base; additional hand view for PegInsertionSide, PlugCharger, StackCube	128×128
LIBERO-PRO	Agent view, wrist	800×512
robosuite	Agent view, wrist	512×512
BEHAVIOR-1K	Ego, left/right wrist	512×512

CALVIN: interaction versus feedback control. The execution analysis reports Astra's average consecutive goals decreasing from 3.10 to 1.10, Sol's from 1.40 to 0.50, and Fable's increasing from 0.90 to 2.90. Astra's median action-interface calls fall from 1,795 to 480.5, and Sol's from 1,247.5 to 734. Their median cumulative prompt tokens rise from approximately 321k and 363k to 1,206k and 1,217k. These totals accumulate prompt tokens over successive requests. Under RGB-expanded, five Astra and eight Sol episodes exhaust the shared per-case token

Table A36. Native execution-step limits shared by both conditions. Values show environment defaults → effective evaluation limits; control frequency is unchanged.

Source	Base → effective	Counter and termination semantics
CLIPort	5 → 10	Pick-and-place actions
RoboCasa/365	1000 → 2000	Control steps; field does not truncate episodes in the current wrapper
RoboTwin 2.0	400 → 800	Native action calls, not all simulator steps
RoboWits	800 → 1600	Control steps for the checked case
ManiSkill	50 → 100	Control steps; PickCube termination/boundary checks verify the effective limit
LIBERO-PRO	4000 → 8000	Wrapper/control counter; field does not truncate episodes
robosuite	1500 → 3000	Wrapper/control counter; field does not truncate episodes
BEHAVIOR-1K	500 → 1000	Environment control steps; later collision revision retains 1000
CALVIN, RLBench, VIMA	No finite limit	Checked instances remain bounded by rounds, calls, tokens, and time
VLABench	Unlimited	Simulation-time limit remains unlimited

allowance. Sol nevertheless increases full-sequence success from zero to one, illustrating why mean progress and completion need separate reporting.

Figure A2a connects these aggregate changes to controller construction and final goal completion in sequence 0557.

Budget utilization and interface effects. Three of Fable's four RGB-expanded CALVIN successes finish at rounds 22, 24, and 25, within the shared execution allowance. These episodes use later rounds to continue control after establishing feedback. Figure A2b examines controller construction in PickCube. Tool semantics also matter: the RGB tool compares colors on a 0–255 scale, whereas some Astra/Sol calls use 0–1 values and tight tolerances, producing empty detections. Fable obtains the red region with [255,0,0] and tolerance 80. Correct units determine whether the controller receives a usable target.

Input and execution audit. The audit covers 7,378 Astra/Sol/Fable requests from the ten ordinary benchmark groups: all contain RGB, archived Astra/Sol responses have no length termination, and Fable has 396. RLBench still contains four Astra, two Sol, and four Fable code-execution timeouts. Astra/Sol failures therefore occur even with delivered images and complete responses; code timeouts and interface semantics provide additional execution-level explanations. VIMA improves for all three agents, and Astra improves on RoboTwin while remaining unchanged over the common 153 tasks. The gains vary with the task and how each model uses the additional visual observations.

E.2 Fixed-Input Multi-Round Review

Multi-round review lets agents revisit their initial answers using the same task observations and scoring criteria. Self-review adds reconsideration, Crop review adds image crops, and Tool-assisted review adds specialist visual and geometric measurements. The three conditions permit up to two, two, and six additional responses, respectively, across Geometry, Spatial Reasoning, and Affordance. Table 3a reports results; sampling and tools are specified below.

Shared cases. Astra and Qwen-27B share 46 cases: 19 Geometry, 18 Spatial Reasoning, and nine Affordance. A 72-case pilot covers 24 benchmark/task strata selected by task category and SHA-256 ordering with seed 20260923, independently of scores and answers. It excludes the 60 exact cases from an earlier performance-selected study. One case per stratum forms a 24-case development set; withholding two more cases that share development scenes leaves the 46 evaluation cases. Some scenes recur, and some cases were inspected during development.

Evidence construction. Tool-assisted review adapts GCA's measurement and reference-frame formalization (Chen et al., 2025b) and SpatialClaw's inspectable execution (Cho et al., 2026). It

Table A37. Review conditions for the same cases and native scoring interfaces. Budgets are additional to native execution; response limits count output tokens.

Condition	Evidence available during review	Additional response budget
Initial	Original task observations	None
Self-review	Original observations and initial answer	Two responses, up to 1,800 tokens each
Crop review	Self-review evidence plus image crops	Two responses, up to 1,800 tokens each
Tool-assisted	Original RGB, crops, and specialist measurements	Six responses, up to 4,096 tokens each; up to 16 specialist requests

formulates the measurement before consulting the candidate answer, then uses a persistent Python session. SAM 3 (Carion et al., 2025) supplies regions; SuperPoint (DeTone et al., 2018) and LightGlue (Lindenberger et al., 2023) supply correspondences; GeoCalib (Veicht et al., 2024) estimates calibration. Numerical routines compute relations and estimates. Finite-value and coordinate checks validate the computed outputs. Remaining semantic errors include selecting a whole-object mask for a functional part and ambiguous translation scale.

Observation and execution controls. Specialist tools receive only previously exposed RGB, without predicted depth, learned 3D reconstruction, private calibration, scoring annotations, or extra video frames. Public depth or masks originally supplied by a task remain available to the base model. Review reuses native candidates where available; one Astra Spatial case starts from its public observation because the candidate is missing. Temperature is zero, and total-token and wall-time accounting includes native execution. Incomplete runs may be retried; the first completed result is retained regardless of correctness, yielding one outcome per condition and case.

Outcome interpretation. Geometry counts improved, tied, and worsened results across 24 metric entries from 19 cases; one Qwen-27B crop entry is unavailable. Spatial accuracy retains 18 assigned cases, with missing answers counted as incorrect. Affordance reports box–box IoU and success at IoU≥ 0.5 over nine cases.

E.3 Native Capability and Complete Task Execution

The main study asks how far a VLM's own perception, reasoning, and action selection support embodied generalism through source-native interfaces. Its central difficulty is completing the task with sufficient precision before the execution budget runs out. Related robot-agent studies help distinguish contextual guidance and supplied control competence from the decisions the model must make itself.

Context and control competence. Robotic in-context learning connects demonstrations and interaction to execution through context-conditioned policies, geometric transfer, world models, or skill- and agent-based control (Huang et al., 2026a). Keeping model weights fixed therefore leaves the source of execution competence to be specified. Learned skill selection (Ahn et al., 2022), programmatic composition of perception and control APIs (Liang et al., 2022), and planning through spatial value maps (Huang et al., 2023a) assign different responsibilities to the model and its executor. GPT-Policy (Cheng et al., 2026) uses demonstrations, action references, and feedback to condition a fixed VLM, while a constrained controller checks and executes its requests. Our baseline retains source-native helpers but leaves task interpretation and action selection to the model; additional specialist modules beyond those helpers enter

only the supplementary review study. This identifies which parts of execution test the model's native capability.

Task composition. Composite tasks require several operations while preserving achieved conditions. Our RoboCasa sample contains seven atomic and three composite tasks; RoboCasa365 contains three atomic and seven composite tasks (Appendix A). PyRUA-Lean (Si et al., 2026) evaluates 18 atomic and 32 composite RoboCasa365 task types at five seeds each, reporting 86.7% and 39.4% success. Task composition changes the completion challenge, and its Target50 tasks differ from our pretrain-designated instances.

External execution support. Learned VLA policies supply sensorimotor competence in addition to model-directed reasoning. Removing PyRUA-Lean's frozen RLDX-1 policy lowers atomic and composite success to 65.6% and 20.0%, respectively (Si et al., 2026). The remaining successes retain programmatic composition of other primitives. Our RoboCasa protocols allow 24 model rounds and 60 minutes of trial time; PyRUA-Lean allows 40 calls for atomic tasks, 100 for composite tasks, and a two-hour episode watchdog. Both support multiple actions per Python response. These results depend on task instances, controller support, and execution allowances as well as the underlying VLM.

Extended interaction without completion. GPT-Policy (Cheng et al., 2026) reports 0/3 successful towel pickups without demonstrations, averaging 96.3 decisions and 24.6 minutes. Human video raises success to 2/3 while reducing these averages to 76.7 decisions and 18.9 minutes; averages include failed trials. Our trajectories expose a related interaction burden: five Astra and eight Sol RGB-expanded CALVIN episodes exhaust their token allowance, while three of Fable's four successes finish at rounds 22, 24, and 25 (Appendix E.1). In PickCube, Astra and Sol use 25 and 32 responses without success, whereas Fable succeeds through calibrated feedback control (Figure A2). Our round and token counts establish budget use; they do not substitute for measured elapsed time. Extended interaction alone does not establish successful task completion.

Precision and terminal goal satisfaction. More interaction also need not resolve the geometric conditions that determine success. In GPT-Policy's plug task, robot video yields 0/3 successes, while adding aligned action references yields 2/3; success requires the plug to remain fully seated after release (Cheng et al., 2026). In our arena, initiation or arrival errors occur in 36 of Astra's 37 failed traces despite its lowest trajectory RMSE (Figure 4). Five HumanCLAW episodes reach the seat and issue the sitting action without establishing the required sitting state (Appendix D.4). These findings make the limit concrete: native VLM competence can produce plausible local decisions and repeated actions while still falling short of precise, complete execution within a finite budget. Embodied generalism requires efficient control, accurate endpoints, and preservation of every required goal condition through termination.

F Case Studies

From spatial judgments to usable contacts. Figure A1 illustrates Section 4.3 with six cases from the first evaluation run.

① *Spatial references.* In these runs, Astra correctly judges placement compatibility but misidentifies object-facing direction and camera translation. ② *Contact versus coverage.* Astra identifies usable tool and furniture regions. For pulling, three models place valid contacts on the handle, but their boxes cover only a fraction of the reference door region.

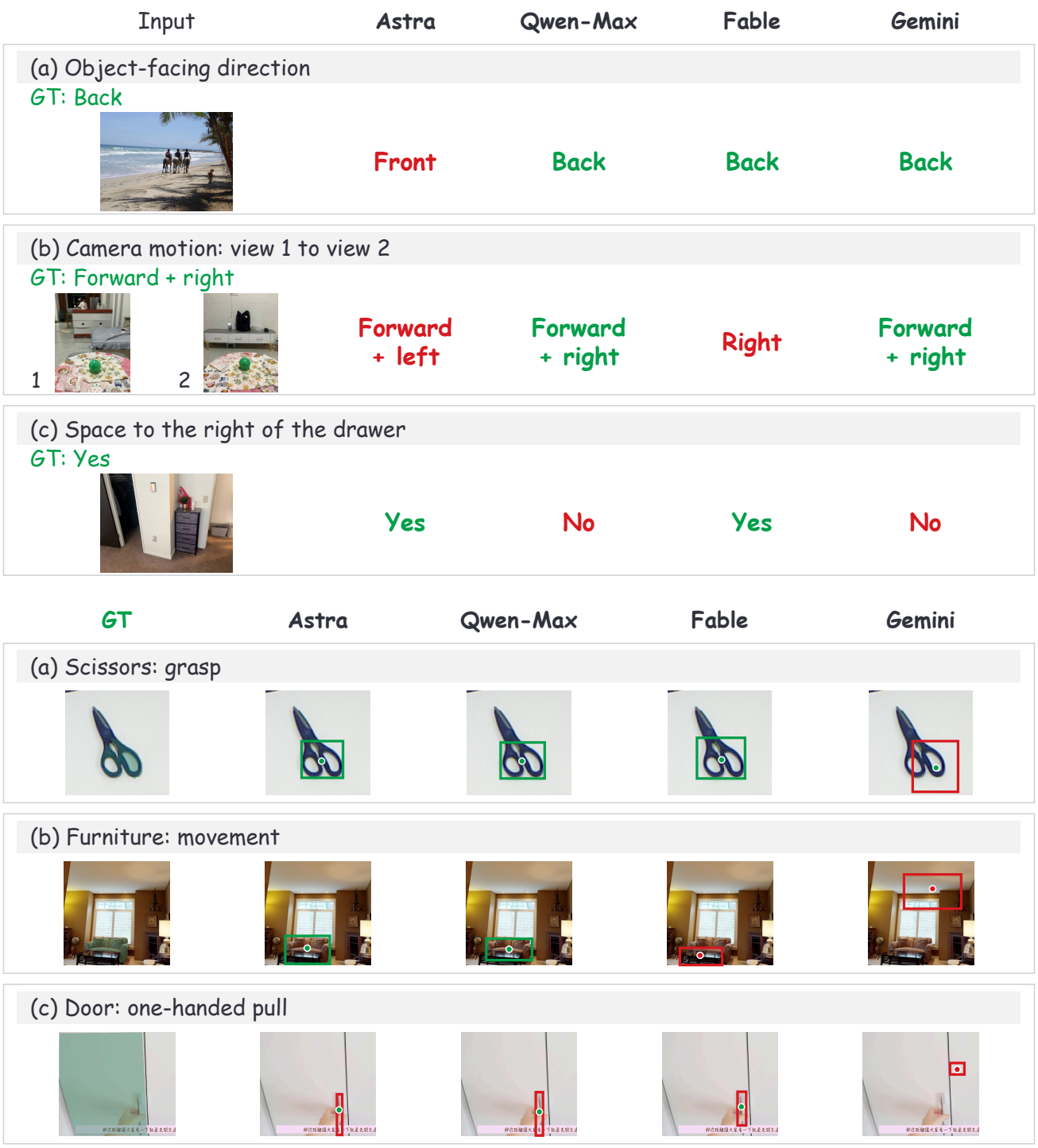


Figure A1. Spatial judgments from the first evaluation run (top: 3DSRBench, MindCube, RoboSpatial) and affordance localization (bottom: UMD, ReasonAff, RAGNet). Top questions concern the dog's facing direction, camera motion, and whether the plant fits beside the drawer; green answers match GT. Below, green points lie inside the GT mask and green boxes have IoU≥ 0.5; red denotes failure.

Controller construction across observation settings. Figure A2 links controller construction to terminal outcomes in CALVIN sequence 0557 and ManiSkill PickCube; execution settings are in Appendix E.1.

CALVIN: Astra's feedback controller emerges later. Baseline Astra builds control from tool-center-point error at round 5; RGB-expanded Astra builds it at round 12. Fable constructs a state-reading movement loop at round 2. Earlier controller construction coincides with longer completed goal sequences in these cases.

PickCube: fewer steps, better organized control. Fable calibrates joint response before constructing reusable numerical feedback control, then succeeds in 36 native steps. Astra and Sol use 100 and 56 steps across 25 and 32 responses without success; their trajectories emphasize direct joint increments and cross-round queries.

Motion and task completion. Figure A3 contrasts visible motion with verified task completion.

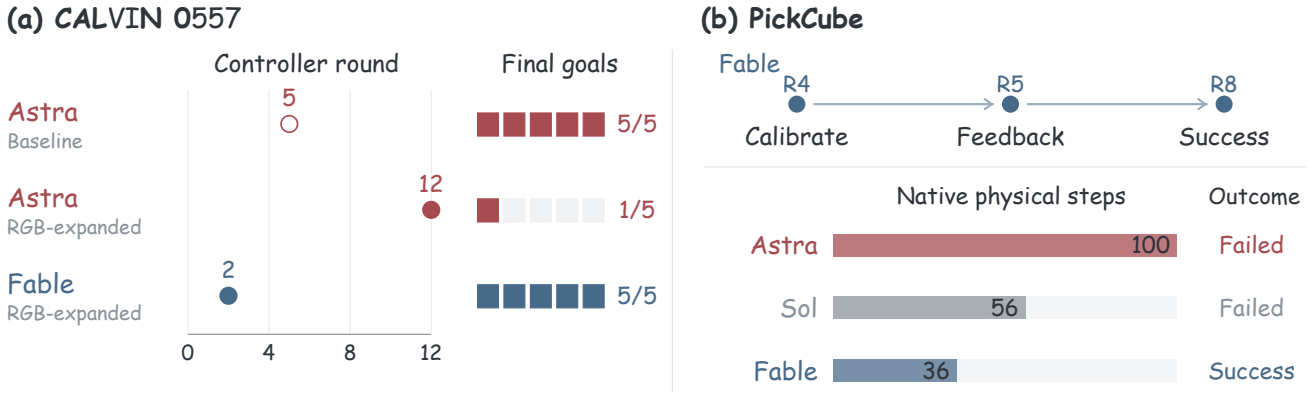


Figure A2. Controller construction and task outcomes. (a) CALVIN dots mark controller construction rounds; filled tiles count final consecutive goals. (b) PickCube shows Fable's ordered calibration, feedback-control, and success events, plus all three agents' physical-step totals and outcomes under RGB-expanded execution. R denotes model rounds; physical steps count environment updates.

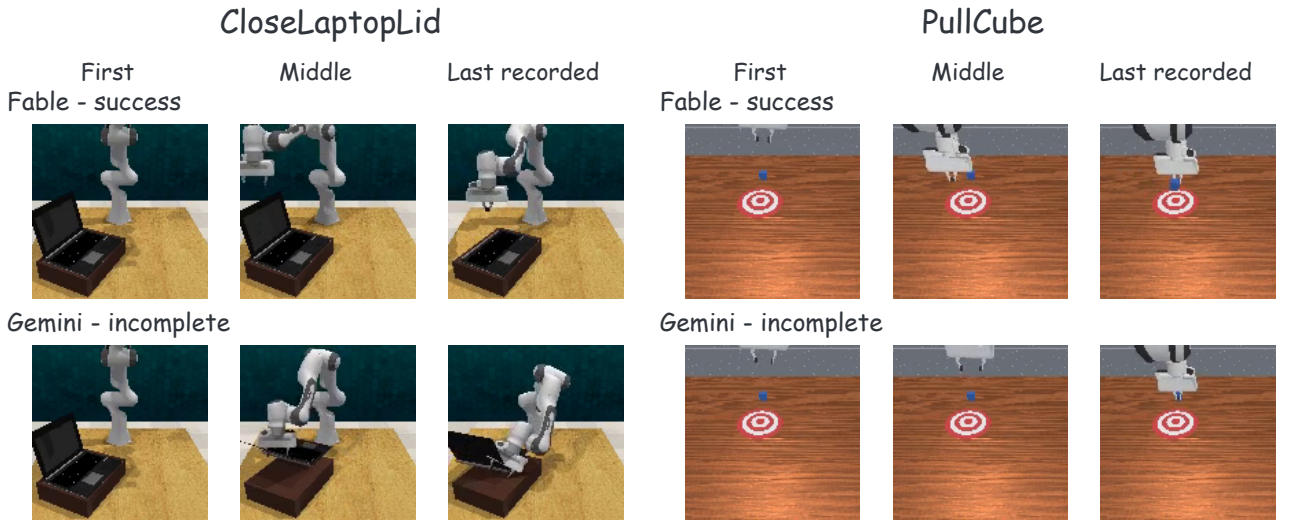


Figure A3. Archived execution contrasts under matched task settings and budgets. CloseLaptopLid uses RLBench's front camera; PullCube uses ManiSkill's base camera. Native records determine success. These earlier episodes are separate from the protocol comparison.

G Evaluation Scales and Public Resources

Table A38 expands Table 1 with instance counts and released artifacts. Task categories, training demonstrations, and source datasets are excluded from the instance count. Our release provides a fixed 1,000-case index across five capability domains, together with observations, episode initializations, and evaluation code.

OpenEQA adds navigation to episodic QA in its active track; Memory denotes cross-session persistence. VLABench's non-interactive track scores action sequences from fixed multi-view images against reference graphs. Hard Vision, Easy Vision varies reasoning effort and tool use; its 55 source benchmarks do not specify an instance count.

Table A38. Reported evaluation populations and released resources. Counts follow the cited papers; artifact availability is checked on 3–4 October 2026. Task-type and source counts describe coverage rather than tested instances. Links lead to author-maintained resources.

Study	Reported evaluation population	Released evaluation resources
OpenEQA (2024)	1,636 QA entries; the 557 HM3D questions also support active navigation. GPT-4V's active evaluation uses a 184-question subset.	Baseline and scoring code; QA manifest and episode-history download instructions.
BEAR (2025)	4,469 multimodal QA entries; robot online validation is additional.	Agent code; benchmark data and scoring scripts.
EmbodiedBench (2025a)	1,128 testing instances in four environments.	Code and configurations; benchmark datasets.
EmbodiedEval (2025)	328 tasks in 125 scenes.	Code; 328-entry test set and scene assets.
Embodied Arena (2025)	22 source benchmarks; a combined evaluated-instance count is not reported.	Platform and configurations; source datasets are downloaded separately.
CaP-X (2026)	Core: seven task types, each with 100 trials per tier. Extensions include 30 LIBERO-PRO task types and two BEHAVIOR tasks with 25 trials each.	Code, environments, and evaluation scripts. The repository lists 39 task types; the paper describes 187 platform tasks.
VLABench (2024)	100 task categories: 60 primitive and 40 composite; episode counts depend on the experiment.	Code and evaluation tracks; fixed-image VLM evaluation data.
WorldLines (2026b)	310 memory QA entries and 21 planning cases; the 62-entry ablation uses a QA subset.	The paper supplies prompts and examples; an author code/data release was not located.
Hard/Easy Vision (2026)	55 benchmarks, 34 capabilities, nine areas; total and per-benchmark evaluated-instance counts are not reported.	Results and qualitative examples; the repository contains a README and overview image, without evaluation scripts or a test manifest.
Embodied Agent Arena	**1,000 cases** from 32 established sources and GeoProbe, including 168 GeoProbe cases.	Harness and evaluation code; case index, observations, and episode initializations.

H Additional Cases Across the Five Capabilities

The examples in Figures A4–A8 connect an explicit task instruction to the model's output or recorded execution. They cover geometric estimation, spatial judgments, functional localization, and complete task outcomes. The current public online case atlas contains 129 selected tasks across 43 benchmark entries and subsets, together with 19 eight-model execution showcases. The latter provide 152 recordings across six planning and thirteen manipulation sources. Selection emphasizes task variety, differing answers, and both successful and unsuccessful runs; these examples do not estimate aggregate accuracy.

Models and execution settings. DeepSeek-1017 cases add first-run predictions on the same static task IDs and recorded ALFRED/CALVIN episodes. Full identifiers and evaluation settings appear in Appendix B.2. The earlier additional material includes Claude Opus 5.5 (recorded identifier `claude-opus-5-5`), now also represented in the updated quantitative snapshot. Selected cases illustrate individual records and do not replace the assigned-case denominators in Table 2. Geometry, Spatial Reasoning, and Affordance panels show each model's first recorded answer; the spatial panels use the first run without YOLOE, excluding later perception-assisted repetitions. In the execution panels, Opus 5.5 uses the recorded source interfaces: ALFWorld visual primitives, DiscoveryWorld text/JSON operations, HumanCLAW egocentric RGB actions, and source-native Manipulation primitives. The saved Manipulation records identify the execution interface but do not establish whether images were supplied at every model round; they are therefore not assigned to the paired observation comparison. Table 2 separately reports the Baseline aggregate. These interfaces may expose source-specific state or helper operations; the examples therefore do not establish a controlled model ranking or an improvement attributable solely to the model.

Reading the evidence. For static tasks, the reference and individual predictions are shown together. Affordance contact denotes membership in the reference mask; box IoU compares the predicted and reference rectangles. These earlier point/box examples are separate from the direct-polygon experiment in Appendix C.5. For execution tasks, the outcome follows the recorded termination and task feedback, rather than an inference from visible motion. Four earlier HumanCLAW recordings with conflicting labels and terminal feedback are excluded from these qualitative comparisons. The quantitative Opus 5.5 results instead use the final 13 completed records in Appendix C.3. The separate Qwen-397B proximity demonstration uses a 20 cm criterion and is also excluded from evaluation. Planning panels contain ordered saved observations; the DiscoveryWorld images visualize the environment rather than the agent's text/JSON input. Manipulation panels sample one recorded camera stream; their timestamps index encoded video, not wall-clock execution. Elapsed runtime is reported separately. A failed terminal state and exhaustion of a model-round limit are distinct stopping conditions.

The cases expose a common gap between partial competence and task completion. Correctly estimating one displacement does not ensure a correct camera-frame transformation; locating a contact does not ensure adequate region coverage; and visible motion does not establish completion of every task predicate. The execution records make this distinction concrete: a HumanCLAW episode ends unsuccessfully before its time ceiling, whereas the Opus 5.5 failures in Figure A8 reach their model-round limits.

(a) GeoProbe | Real Object Motion

Track the green-handled glass jug relative to the table.

View 1 View 2

Model	Direction / cm
GPT-6 Astra	right / 2
GPT-6 Sol	right / 2
Fable 5.1	right / 4
Gemini 3.8 Flash	none / 0
Qwen 3.8 Max	right / 3
Qwen 3.5 397B	none / 0
Qwen 3.5 27B	left / 7
Opus 5.5	right / 2.5
DeepSeek-1017	up-right / 4

Reference: right, 2 cm

(b) GeoProbe | Camera Translation

Estimate camera B's position in the coordinate frame of camera A.

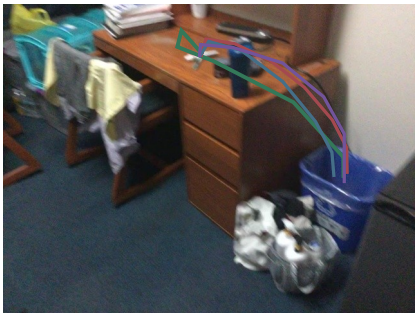
View 1 View 2

Model	Translation (cm)
GPT-6 Astra	[8, 0, -10]
GPT-6 Sol	[10, -10, 0]
Fable 5.1	[6, -3, 0]
Gemini 3.8 Flash	[25, 12, 0]
Qwen 3.8 Max	[10, 4, 0]
Qwen 3.5 397B	[15, 8, 5]
Qwen 3.5 27B	[8, 3, -12]
Opus 5.5	[15, -5, 0]
DeepSeek-1017	[6, -5, 12]

Reference translation: [12, 9, 0] cm

(c) TraceSpatial | 3D Trajectory

Predict a trajectory from the toothpaste to the blue waste bin.



GT Astra Fable Opus 5.5

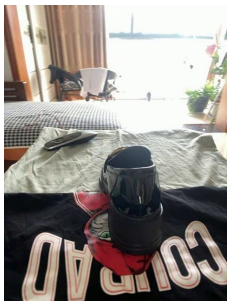
Image projection of 3D paths

Model	3D Fréchet (m)
GPT-6 Astra	0.525
GPT-6 Sol	0.634
Fable 5.1	0.826
Gemini 3.8 Flash	1.148
Qwen 3.8 Max	0.699
Qwen 3.5 397B	0.715
Qwen 3.5 27B	0.497
Opus 5.5	0.549
DeepSeek-1017	0.556

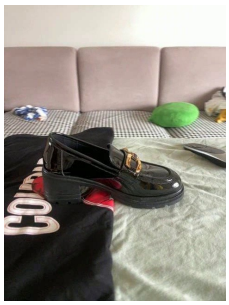
Figure A4. Geometry: displacement, reference frames, and trajectories. First-run predictions for three selected cases. (a) The jug moves right by 2 cm relative to the table; predictions distinguish movement, direction, and distance. (b) Camera translation is expressed in camera A's frame, with reference $[12, 9, 0]$ cm; orientation is omitted from this panel. (c) TraceSpatial paths are projected into the image, while the reported Fréchet distance is measured in 3D meters, distinct from aggregate trajectory RMSE. The reference and Astra, Fable, and Opus 5.5 paths are drawn; all nine individual errors are listed. These cases probe different geometric requirements, rather than interchangeable measures of task success.

(a) MindCube | Viewpoint Change

In which direction did the camera move between these views?



View 1



View 2

Model	First-run answer
GPT-6 Astra	D
GPT-6 Sol	B
Fable 5.1	A
Gemini 3.8 Flash	A
Qwen 3.8 Max	A
Qwen 3.5 397B	C
Qwen 3.5 27B	A
Opus 5.5	A
DeepSeek-1017	A

Reference A: diagonally forward and right

(b) MMSI-Bench | Relative Direction

If the open cupboard faces north, where is the cooking area?

View 1

View 2

Model	First-run answer
GPT-6 Astra	A
GPT-6 Sol	A
Fable 5.1	A
Gemini 3.8 Flash	D
Qwen 3.8 Max	C
Qwen 3.5 397B	B
Qwen 3.5 27B	B
Opus 5.5	D
DeepSeek-1017	D

Reference A: northwest

(c) 3DSRBench | Object Orientation

Which side of the person is facing the camera?

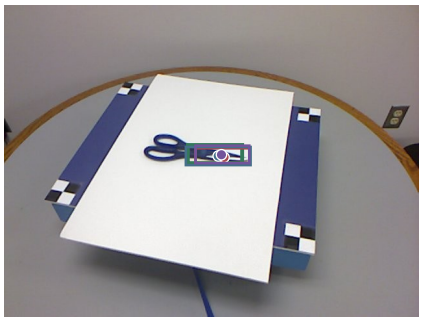
Model	First-run answer
GPT-6 Astra	B
GPT-6 Sol	B
Fable 5.1	C
Gemini 3.8 Flash	B
Qwen 3.8 Max	B
Qwen 3.5 397B	C
Qwen 3.5 27B	B
Opus 5.5	C
DeepSeek-1017	C

Reference B: left

Figure A5. Spatial Reasoning: errors depend on the reference frame. Selected first-run answers without YOLOE. (a) MindCube asks about camera movement between two views: A is diagonally forward-right, B right, C left, and D diagonally forward-left. (b) MMSI-Bench displays two selected views from its multi-view input and asks for a direction anchored to the open cupboard: A is northwest, B southwest, C north, and D northeast. (c) 3DSRBench asks which side of the person faces the camera: A is front, B left, C back, and D right. Green letters match the source reference; red letters do not. Different agents succeed on different frame transformations.

(a) UMD | Cutting

Locate the cutting region of the scissors.



GT box Astra Fable Opus 5.5

Reference box and first-run predictions

Model	Contact / box IoU
GPT-6 Astra	inside / 0.636
GPT-6 Sol	inside / 0.694
Fable 5.1	inside / 0.688
Gemini 3.8 Flash	outside / 0.000
Qwen 3.8 Max	inside / 0.706
Qwen 3.5 397B	inside / 0.540
Qwen 3.5 27B	outside / 0.000
Opus 5.5	inside / 0.734
DeepSeek-1017	outside / 0.406

(b) ReasonAff | Pulling

Locate the functional region to pull.

GT box Astra Fable Opus 5.5

Reference box and first-run predictions

Model	Contact / box IoU
GPT-6 Astra	inside / 0.197
GPT-6 Sol	inside / 0.150
Fable 5.1	inside / 0.103
Gemini 3.8 Flash	outside / 0.000
Qwen 3.8 Max	inside / 0.193
Qwen 3.5 397B	outside / 0.000
Qwen 3.5 27B	outside / 0.000
Opus 5.5	inside / 0.156
DeepSeek-1017	inside / 0.148

(c) RAGNet-3DOI | Two-Hand Interaction

Locate the region for the task “free using two hands”.

GT box Astra Fable Opus 5.5

Reference box and first-run predictions

Model	Contact / box IoU
GPT-6 Astra	outside / 0.000
GPT-6 Sol	outside / 0.000
Fable 5.1	inside / 0.906
Gemini 3.8 Flash	outside / 0.045
Qwen 3.8 Max	outside / 0.000
Qwen 3.5 397B	outside / 0.000
Qwen 3.5 27B	outside / 0.000
Opus 5.5	inside / 0.942
DeepSeek-1017	outside / 0.000

Figure A6. Affordance: usable contact and region coverage can disagree. First-run outputs on UMD, ReasonAff, and RAGNet-3DOI. Green rectangles are reference boxes; colored points and boxes show Astra, Fable, and Opus 5.5 predictions. The table includes all nine models: contact is inside/outside the reference mask, and box IoU is intersection over union with the reference rectangle. Out-of-image points count as outside. The scissors and pulling examples contrast contact localization with region coverage; the two-hand example illustrates disagreement over the relevant functional region. Short task descriptions paraphrase the input instructions without changing the target.

(a) ALFWorld | Object and Light

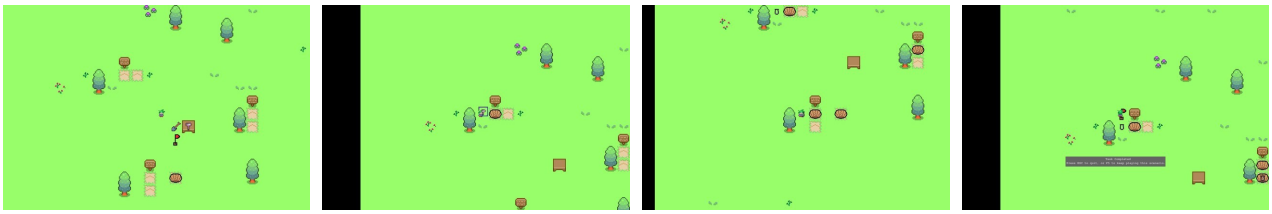
Turn on a lamp while holding the requested object.

obs. 1 obs. 3 obs. 6 obs. 8

Claude Opus 5.5 Reported success | 8 code rounds | 80.9 s

(b) DiscoveryWorld | Archaeology

Date the artifacts and flag the site containing the oldest one.

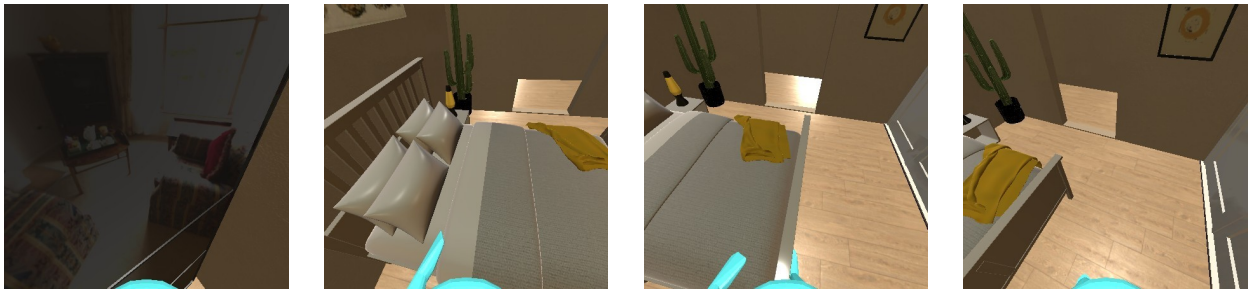


obs. 1 obs. 13 obs. 31 obs. 44

Claude Opus 5.5 Reported success | 7 code rounds | 167.9 s

(c) HumanCLAW | Sitting

Find the couch, approach until touching it, and sit.



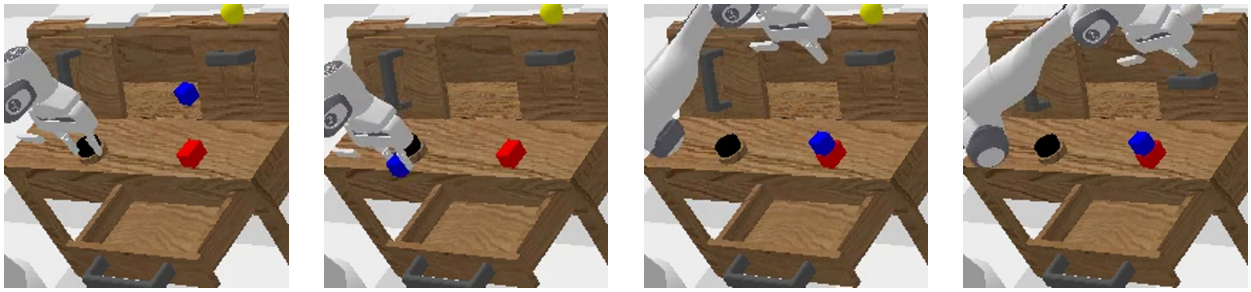
obs. 1 obs. 20 obs. 49 obs. 69

Claude Opus 5.5 Task failure | 70 code rounds | 933.6 s

Figure A7. Task Planning: a terminal episode need not be a completed task. Additional Opus 5.5 runs. (a) ALFWorld reports success after 8 code rounds under ceilings of 20 minutes, 8 rounds, 120 environment steps, 540 primitive calls, and 320k tokens. (b) DiscoveryWorld reports success after 7 rounds under 20 minutes, 8 rounds, 100 steps, 400 calls, and 320k tokens; its text/JSON interaction is illustrated with environment snapshots. (c) HumanCLAW stops after 70 rounds (933.6 s) without completing the couch-sitting task, under ceilings of 180 minutes, 101 rounds, 100 steps, 400 calls, and 2M tokens. The last case is a failed terminal state, not a wall-clock timeout. Frames are sampled saved observations, not continuous video.

(a) CALVIN | Sequence 0526

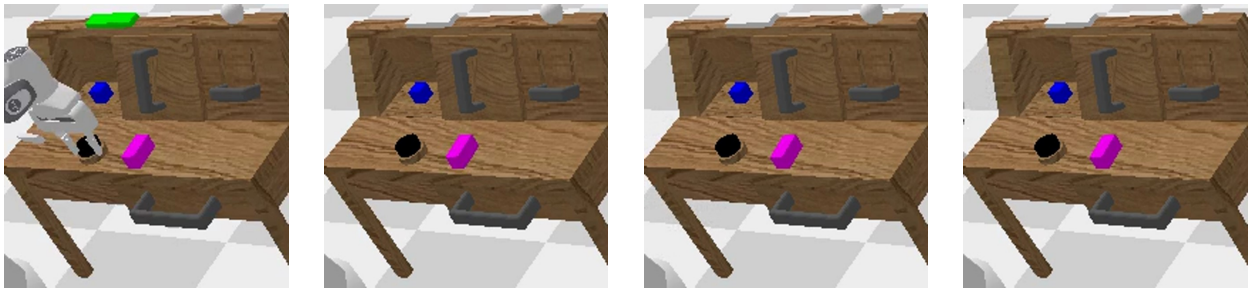
Complete the five instructed manipulation subgoals in sequence.



Claude Opus 5.5 Reported success | 40-round ceiling | 229.4 s

(b) CALVIN | Sequence 0557

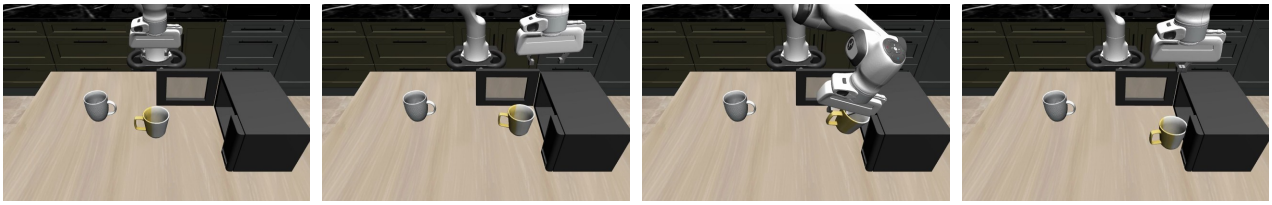
Complete the five instructed manipulation subgoals in sequence.



Claude Opus 5.5 Round limit reached | 40 rounds | 978.6 s

(c) LIBERO-PRO | Mug and Microwave

Put the yellow-and-white mug in the microwave and close it.



Claude Opus 5.5 Round limit reached | 24 rounds | 1308.6 s

Figure A8. Manipulation: motion can continue without completing all goals. Additional Opus 5.5 runs through recorded source-native interfaces. (a) CALVIN sequence 0526 reports success; (b) sequence 0557 exhausts 40 model rounds. Both allow 7,000 primitive calls and 1.28M tokens. (c) LIBERO-PRO requires placing a mug inside the microwave and closing it; the run exhausts 24 model rounds, with ceilings of 256 primitive calls and 640k tokens. Outcomes come from the terminal reports. Frame labels index the encoded recording, whereas the seconds beside each outcome are wall-clock runtime. Different task instances and source interfaces are illustrated here, not a matched comparison of models.

H.1 Execution Across Models

Figures A9 and A10 pair task instructions and chronological frames with outcomes, model rounds, and elapsed runtime for all nine agents. ALFRED covers household goals; CALVIN requires five ordered manipulation subgoals. Tasks are selected separately for each model. DeepSeek-1017 shares the spray-bottle task with Astra and the CALVIN 0526 sequence with five other agents.

Completion, interaction, and time. Table A39 groups identical task identifiers. Fable completes the RoboCasa drawer task in 20 rounds; four other agents exhaust their budgets at 24, spanning 215.5–3,223.2 s. On LIBERO-PRO, Qwen-397B exhausts its budget while three agents complete the placement. On ALFWorld, Sol succeeds in fewer rounds than Fable (6 versus 7), but takes more than twice as long (433.8 versus 197.5 s). Thus, neither interaction count nor elapsed time alone establishes complete, timely task execution.

Execution settings. ALFRED and ALFWorld use recorded Baseline interfaces. DeepSeek-1017 CALVIN uses RGB-expanded observations with the source-specific ceilings in Table A16. CALVIN's original seven agents use RGB-expanded; the Opus 5.5, LIBERO-PRO, and RoboCasa records identify native primitives without a verified per-round image schedule. These episodes remain separate from Table 2's Baseline aggregate and the paired observation experiment. Runtime includes model and tool execution, not hardware-normalized inference alone. Budget exhaustion need not coincide with the model-round ceiling; the recorded termination is retained without attributing it to an unverified limit.

Table A39. Shared-task execution records. Rows within each block use the same task identifier and source interface. Rounds count recorded model responses; time is elapsed wall-clock runtime. ALFWorld allows 8 rounds, 120 environment steps, and 20 minutes. LIBERO-PRO and RoboCasa allow 24 rounds and 256 primitive calls, with token ceilings of 640k and 1.12M, respectively. These are selected episodes, not repeated-run averages.

Model	Rounds	Time (s)	Recorded outcome
ALFWorld: put a cleaned bar of soap on a counter			
GPT-6 Sol	6	433.8	Success
Claude Fable 5.1	7	197.5	Success
LIBERO-PRO: put the tomato sauce in the basket			
Claude Fable 5.1	16	1,553.5	Success
Gemini 3.8 Flash	15	429.0	Success
Qwen 3.5 397B-A17B	24	934.6	Budget exhausted
Claude Opus 5.5	20	2,376.8	Success
RoboCasa: open the left drawer			
GPT-6 Sol	24	735.5	Budget exhausted
Claude Fable 5.1	20	618.1	Success
Gemini 3.8 Flash	24	246.3	Budget exhausted
Qwen 3.8 Max	24	3,223.2	Budget exhausted
Qwen 3.5 27B	24	215.5	Budget exhausted

Task Planning | ALFRED

Selected episodes; first and last displayed frames; outcome from the terminal report.

(a) GPT-6 Astra

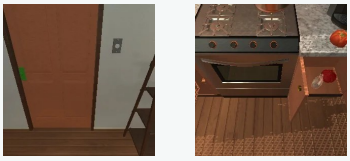
Put the green spray bottles on the toilet.

obs. 1 obs. 7

Success
8 rounds | 153.2 s

(b) GPT-6 Sol

Place a salt shaker in a pullout drawer.

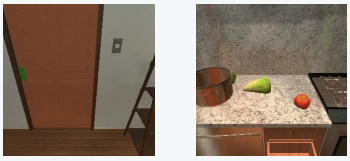


obs. 1 obs. 8

Success
8 rounds | 254.1 s

(c) Claude Fable 5.1

Place a salt shaker in a pullout drawer.



obs. 1 obs. 6

Success
8 rounds | 222.5 s

(d) Gemini 3.8 Flash

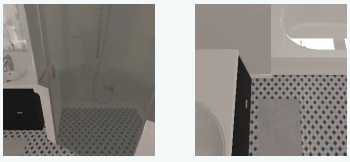
Wash the pink bar by the sink; put it in the tub.

obs. 1 obs. 7

Success
8 rounds | 175.4 s

(e) Qwen 3.8 Max

Wash the pink bar by the sink; put it in the tub.

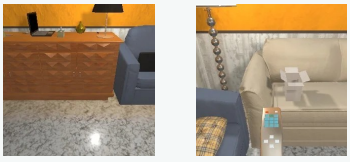


obs. 1 obs. 7

Success
5 rounds | 486.7 s

(f) Qwen 3.5 397B-A17B

Hold a remote and turn on the lamp.

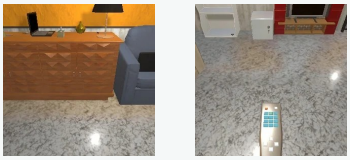


obs. 1 obs. 4

Success
5 rounds | 63.8 s

(g) Qwen 3.5 27B

Hold a remote and turn on the lamp.

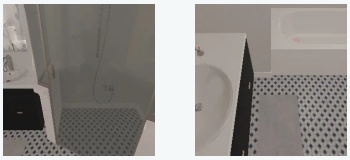


obs. 1 obs. 6

Budget exhausted
8 rounds | 102.1 s

(h) Claude Opus 5.5

Wash the pink bar by the sink; put it in the tub.



obs. 1 obs. 6

Success
8 rounds | 425.8 s

(i) DeepSeek-1017

Put the green spray bottles on the toilet.

obs. 1 obs. 2

Budget exhausted
8 rounds | 71.1 s

Figure A9. Nine-model task execution in ALFRED. Each panel shows the first and last displayed observations from an individual Baseline episode, with a shortened instruction, terminal outcome, rounds, and wall-clock runtime. All runs allow 8 model rounds, 120 environment steps, and 20 minutes. DeepSeek-1017 exhausts 8 rounds on the same spray-bottle task as Astra. The two Qwen models in panels (f,g) share the remote-and-lamp task: Qwen-397B reports success after 5 rounds, whereas Qwen-27B exhausts its budget at 8. Other panels cover separately selected tasks. Saved observations preserve temporal order with unequal intervals.

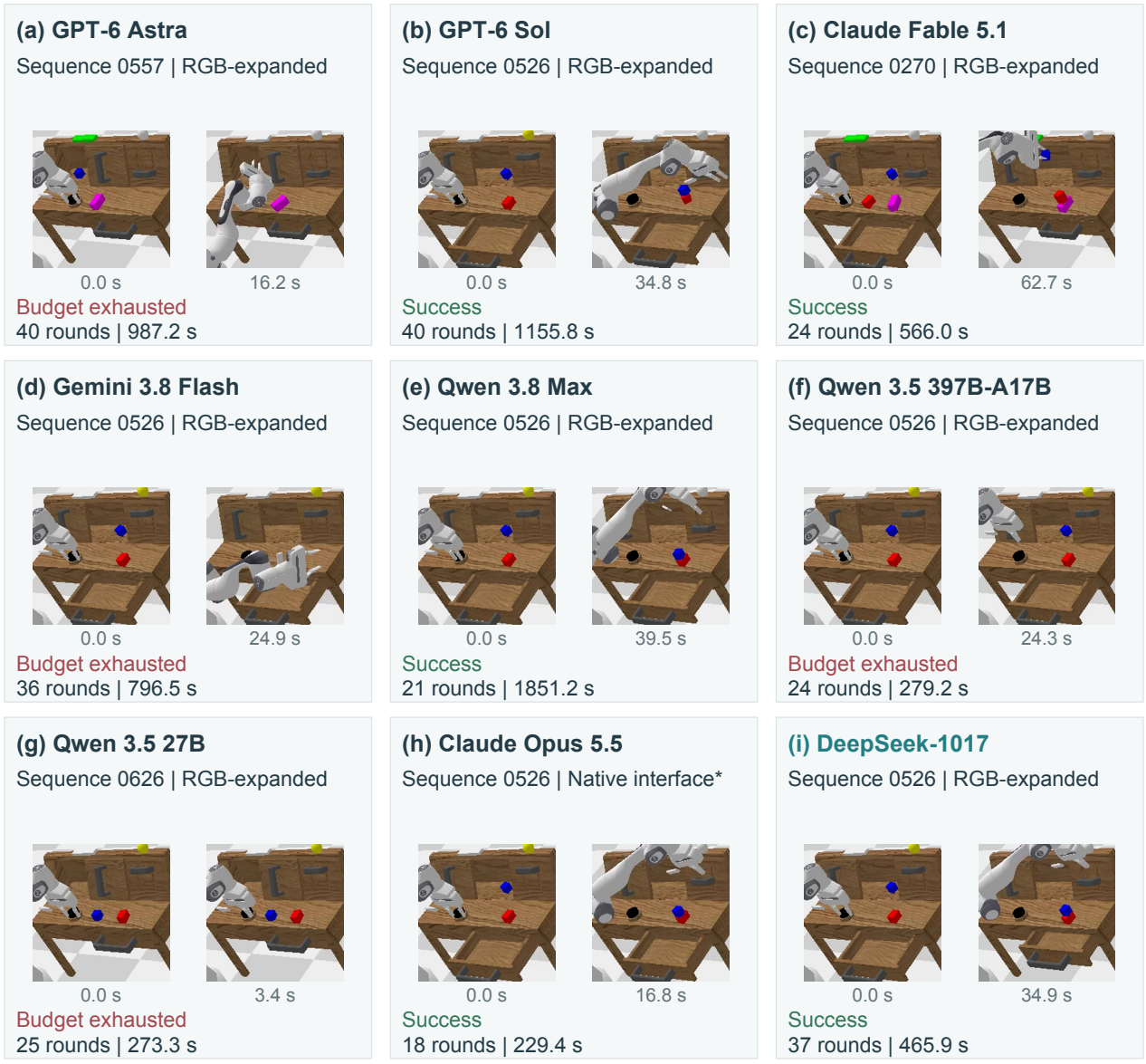


Figure A10. Nine-model execution of CALVIN sequences. Each sequence requires five ordered subgoals. Sequence 0526 requires retrieving the blue block from the sliding cabinet, storing it in the drawer, retrieving and stacking it, and switching off the bulb. Sol, Qwen Max, Opus 5.5, and DeepSeek-1017 report success on this sequence; Gemini and Qwen-397B report budget exhaustion. Each run allows 40 model rounds, 7,000 primitive calls, and 1.28M tokens. Panels (a–g,i) use RGB-expanded; *Opus 5.5 records native primitives without a verified per-round image schedule. DeepSeek-1017 completes the sequence in 37 recorded responses and 465.9 s, within a 120-minute episode ceiling. The other eight selected episodes retain their recorded limits. Each panel shows two selected frames; their labels give encoded-video time, while the outcome line gives wall-clock runtime. Tasks and observation settings vary across panels.